

Day 1 & 2:

Introduction to SD-WAN and Wide Area Networking

1. Course Overview and Promise

The speaker, Vishnu Dutt from Cisco, introduces a comprehensive 24-hour course on SD-WAN. The primary promise of the course is to equip participants with the ability to "design any SDN network from scratch" upon completion. This capability is promised regardless of prior experience, ranging from 2 to 15 years, suggesting a foundational and practical approach.

- **Key Takeaway:** The course aims to make all participants capable of designing and discussing SD-WAN value propositions to customers, regardless of their current experience level.
- **Quote:** "If you complete this complete course... I promise you will be capable of designing any SDN network from scratch. It doesn't matter that if you have an experience of 2 years or 6 years or 10 years or 15 years, you all will be in the same place that you can design, you can talk about SD-WAN proposition value to any customer."

2. Fundamental Importance and Course Structure

A crucial emphasis is placed on understanding the fundamentals of SD-WAN technology. The course dedicates a significant portion, "the initial... 16 hours," to exploring these fundamentals in depth before moving on to practical design and migration topics in the last eight sessions. The speaker notes that most deployments encountered will be "brownfield" (existing infrastructure), rather than "greenfield" (new deployments).

- **Key Takeaway:** Strong foundational knowledge is critical for effective network design. The course prioritizes deep dives into fundamentals before practical application.
- **Quote:** "Believe me, if you want to design a network, the fundamentals of this technology is going to be crucial for you. So that is why in the initial, in the starting, what we can say 16 hours, we are going to look, learn all the fundamentals related to this technology."
- **Fact:** The course includes hands-on demonstrations using vManage and vSmart, although separate videos will be provided for dedicated hands-on exercises.

3. Session 1 Agenda and Approach

Session 1 is designed to start "absolutely from scratch," assuming no prior knowledge. The agenda progresses logically:

- **Defining Wide Area Networking (WAN):** Starting with the basic definition.
- **History of WAN:** Examining the evolution of WAN over the past 20-25 years to understand how SD-WAN adds value.
- **Typical Enterprise Network Analysis:** Discussing the components and infrastructure of a standard enterprise network, particularly its WAN aspect, to identify existing problems.
- **Problem-Solution Framework:** The core idea is that understanding the "problems we have in current enterprise network" is essential before appreciating how SD-WAN provides a solution.
- **Vendor Agnostic Perspective:** Critically, this session avoids specific vendor terminologies (e.g., Cisco, Viptela, Silver Peak) to focus on general principles like "overlay and underlay concept." The goal is to understand *why* SD-WAN is chosen and the methodologies employed across the industry.
- **Puzzle Metaphor:** SD-WAN is presented as a "puzzle" that will be broken into pieces, understood individually, and then reassembled to form a complete solution.
- **Key Takeaway:** The initial learning approach is problem-driven and vendor-agnostic, building a foundational understanding of WAN challenges that SD-WAN addresses.
- **Quote:** "If you understand why we have introduced software-defined wide area networking, you need to understand what are the problems we have in current enterprise network. If you understand the problems, you can think of a solution, and if you're thinking a solution, then SD-WAN should be there in your mind because it is actually solving the problem of the legacy WAN."
- **Quote:** "We are trying to think that SD-WAN is a puzzle. We will break it into multiple pieces. We will try to understand the piece and then at last we are going to put all pieces together to build a complete solution."

4. Understanding Wide Area Networking (WAN) through a Business Case

The speaker introduces a practical business scenario to illustrate the concept of WAN:

- **Scenario:** Mr. Parth opens "ABC Sports," a company manufacturing sporting goods. Initially, his business operates from a single "campus" in Delhi, where employees collaborate, and a public website (abc.com) runs on a server. This single-location setup is referred to as a "campus network."
- **Growth and Expansion:** As ABC Sports grows, demand emerges from other major Indian cities like Mumbai, Bangalore, and Chennai. Mr. Parth decides to open "branches" in these cities.
- **The "Problem":** Employees in these new branches need to access resources (e.g., the file server) located at the Delhi campus. This necessitates connectivity between geographically distant locations.

- **Definition of WAN:** Any technology that connects these "multiple branches and campus together which are far away comes into the area of comes under the umbrella of wide area networking." The example highlights a 1,000 km distance between Mumbai and Delhi to emphasize the "wide area" aspect.
- **Branch-to-Branch Connectivity:** The scenario also considers the possibility of direct connectivity between branches (e.g., Mumbai to Bangalore) or routing all branch traffic through the main campus/data center for cost savings and security, depending on business needs.
- **Role of Service Provider:** The need for WAN connectivity leads Mr. Parth to explore options from a "service provider" who can solve his problem of connecting geographically dispersed locations.
- **Key Takeaway:** WAN is defined as the technology connecting geographically distant enterprise locations (campuses and branches) to facilitate collaboration and resource sharing. Business growth often necessitates the expansion from a local campus network to a wider WAN.
- **Quote:** "Any technology which is right here which is connecting this branch to this campus and believe me the diff- the distance between Mumbai and and Delhi is approximately 1,000 kilometers... any technology which is which is connecting campus with branch that comes into the area of wide area networking."
- **Quote:** "Any technology which is into this cloud and connecting multiple branches and campus together which are far away comes into the area of comes under the umbrella of wide area networking."

Day 3 & 4:

SD-WAN Fundamentals: A Detailed Briefing

1. Introduction to SD-WAN and Learning Methodology

The session serves as a continuation of an "SD-WAN from scratch" course, building upon previous discussions of WAN history and current problems in legacy enterprise networks. The instructor, Vishnu Dut, emphasizes that SD-WAN "is adding value from a single pane of glass you can do many things right it is going to be easy to manage it is going to be easy to deploy your wan infrastructure."

The core pedagogical approach is to "break down the SD wan puzzle" into "multiple pieces together" and then combine them to form a complete solution. This mirrors the development of the TCP/IP model, which "divided the complete problem...into seven – five layers and you and they work on these layer individually and then they combine then basically they got the solution." The three main pieces of the SD-WAN puzzle identified are:

- **Control Plane**
- **Next Hop**
- **Tunneling**

2. Control Plane vs. Data Plane

A significant portion of the session is dedicated to clarifying the distinction between the control plane and data plane, a concept often misunderstood.

- **Control Plane:** This refers to "anything which is helping me to create this routing table." It is the intelligence and decision-making part of the network device.
- **Example (OSPF/EIGRP):** In OSPF, Link State Advertisements (LSAs) received by routers and the LSA database are part of the control plane. Similarly, EIGRP's topology table is part of the control plane. These elements "directly processed by the route processor of the router" and are "CPU intensive."
- **Function:** The control plane is responsible for gathering network topology information, running routing algorithms (like SPF in OSPF), and ultimately building the routing table. Without LSAs/topology tables, "we are never ever going to form the routing table."
- **Data Plane (Forwarding Plane):** This is the actual mechanism that forwards traffic based on the information provided by the control plane.
- **Routing Table:** The routing table itself, which dictates "if you want to go 2.2.2.0/24 you can go via R2," is considered the data plane.
- **Cisco Express Forwarding (CEF) / Forwarding Information Base (FIB):** CEF (or FIB) is an optimized, more granular version of the routing table that includes all necessary information (e.g., next-hop MAC addresses) to "switch the packet" quickly, without needing additional lookups. "Whatever is needed to switch this packet we can find in a single location which is called Cisco express forwarding." The FIB is derived from the routing table; "If we do not have routing table, we cannot think of creating forwarding information base."
- **Function:** When a packet arrives, the data plane consults the routing table (or FIB) to determine the outgoing interface and next hop, then forwards the packet.

- **Separation of Planes:** Traditionally, control and data planes reside within a single router. However, SD-WAN aims to separate these planes. The instructor poses the question: "Can we separate this control plane with data plane? Can my control plane be anywhere else and data plane can be anywhere else?" The benefit of this separation, especially in large networks, is to centralize control and simplify management. Instead of every router forming neighbor relationships with every other router (resulting in $N*(N-1)/2$ relationships for N routers), an "intelligent guy" (a centralized control plane device) can handle all routing information.
- **Centralized Control Plane:** In this model, routers at the edge (data plane) only need to form a neighbor relationship with this "intelligent device" (control plane). This device "knows all the routes in your topology" and pushes routing information down to the edge routers, enabling them to forward data even without direct neighbor relationships among themselves. This concept is likened to BGP route reflectors.

3. Next-Hop Manipulation

The ability to manipulate the next hop is presented as a crucial element for controlling network topologies in SD-WAN.

- **Traditional Routing:** In typical IGP protocols (like OSPF or EIGRP), the next hop for a route is the directly connected router advertising that route.
- **BGP and Next-Hop-Self:** BGP, by default, exhibits a different next-hop behavior, especially within an Autonomous System (AS). The next hop for routes learned via EBGP is often preserved, even when advertised internally via IBGP. This means an internal router might need to reach an external next hop indirectly. BGP includes a "next-hop self" command to change this behavior.
- **SD-WAN Application:** The power of SD-WAN lies in its ability to **dynamically change the next hop** from a centralized point (the "intelligent guy").
- **Use Case: Hub-and-Spoke Topology:** By default, SD-WAN fabrics often establish a full-mesh data plane connectivity between all sites. However, if a boss requires a hub-and-spoke topology (e.g., all traffic from Branch 2 to Branch 3 must first go through Data Center), the "intelligent guy" (control plane) can modify the next hop for routes advertised to the spokes. For instance, Branch 1 would learn that to reach Branch 2's subnet, the next hop is the Data Center, not Branch 2 directly. This central control allows for on-the-fly topology changes without reconfiguring individual edge routers.
- **Core Idea:** "If you can control the next hop, if you can change the next hop and if you can change the next hop, you can have topology of your own choice."

4. Tunneling (Overlay Networks)

Tunneling is the mechanism that creates the "overlay" network, allowing secure and logical connections over an "underlay" (e.g., the Internet).

- **GRE as an Example:** Generic Routing Encapsulation (GRE) tunnels are used as an initial example to illustrate tunneling. GRE creates "an illusion that R1 and R2 are directly connected" over an underlying network (like the internet).
- **Packet Encapsulation:** When traffic needs to traverse a GRE tunnel, the original "inner IP packet" is encapsulated with a GRE header and an "outer IP packet" header. The outer header uses the tunnel endpoint IP addresses (the public IPs of the routers on the underlay), allowing the packet to traverse the underlying internet. The GRE header informs the receiving tunnel endpoint that there's another IP packet inside.
- **Actual Flow:** While it feels like a direct link, the traffic actually flows over the underlay network, encapsulated. "In actual there is no link but basically there is an encapsulation."
- **SD-WAN and Tunnels:** SD-WAN heavily relies on tunneling for both control and data planes.
- **Control Plane Tunnels (Green Links):** These are secure tunnels (e.g., TLS/DTLS based) formed between edge routers and the centralized control plane device ("Mr. Smart"). They are "totally dedicated to only one thing for your control planes," which is sharing routing information.
- **Data Plane Tunnels (Yellow Links):** These are separate tunnels (typically IPsec in Cisco SD-WAN) established between edge routers for actual user data traffic. They are formed "because you want to securely send your data over internet." By default, based on the next-hop information received from the control plane, these tunnels create a full-mesh topology between sites that need to communicate directly.
- **Underlay vs. Overlay:** The underlying physical network (e.g., Internet, MPLS) is the "underlay," providing basic reachability. The logical tunnels built on top are the "overlay," which is "not the actual link. This is giving you an impression that R1 and R2 are directly connected." The instructor emphasizes that "the entire solution whatever is known as SD wan fabric... is just nothing whatever the overlay you are creating is your fabric."

5. Combining the Concepts: The SD-WAN Solution (Pre-Terminology)

The session then integrates these three core concepts (control plane/data plane, next hop, tunneling) into a cohesive, vendor-agnostic SD-WAN solution architecture.

- **Goal:** The primary objective of SD-WAN is that "every route should be known by every other guy," meaning every branch and data center should have full routing awareness of the entire network.

1. **Architecture: Edge Routers (Branches/Data Centers):** When these routers come online and connect to the internet (underlay), they first establish **secure control plane tunnels (green links)** with a central "Mr. Smart" device.
2. **Route Exchange (Control Plane):** A "routing protocol RP" (later revealed as OMP) runs over these green tunnels, allowing edge routers to share their local routes (LAN subnets) with "Mr. Smart."
3. **Centralized Intelligence:** "Mr. Smart" gathers "all the routes that belongs to your company." It then advertises these routes back to all edge routers via the same control plane tunnels. By default, it advertises everything it learns, ensuring full routing information dissemination.
4. **Data Tunnel Formation (Data Plane):** Based on the next-hop information received from "Mr. Smart," edge routers proactively establish **secure data plane tunnels (yellow links)** with each other. These tunnels (typically IPsec) are used for actual user data traffic.
5. **Mr. Smart's Role (Control Plane Only):** Crucially, "Mr. Smart is not coming into picture you are not sending any data to Mr. Smart so it is not the part of your data flow or data traffic flow." Its sole purpose is to manage and distribute routing information.
6. **Topology Manipulation:** The real power emerges when the customer wants to change the default full-mesh data plane topology. By instructing "Mr. Smart" (e.g., "please make my data center as hub and my branches as spokes"), "Mr. Smart" modifies the next-hop advertisements for routes. If a spoke is told that another spoke's route is reachable via the hub, it will no longer form a direct data tunnel with that spoke, forcing traffic through the hub. This central control allows for on-the-fly design changes.

6. Introducing Cisco SD-WAN (Viptela) Terminology

After establishing the fundamental concepts, the session introduces specific Cisco SD-WAN (Viptela) terminology.

- **Edges (vEdge/cEdge):** The routers at the branches and data centers that form tunnels and carry data traffic. These are the "VH" (Viptela Edge) or "CH" (Cisco Edge) routers.
- **vSmart Controller (Mr. Smart):** The centralized control plane device. It manages routing information, builds the network topology, and orchestrates policy application. It's the "brain" that holds all routing information and can be instructed to modify behavior.
- **vManage:** The "single pane of glass" Graphical User Interface (GUI) where administrators log in to manage and configure the entire SD-WAN fabric. It sends instructions to the vSmart controller. This is effectively the **management plane**.
- **Control Connections:** The secure tunnels between edges and the vSmart controller, built using **TLS (Transport Layer Security)** and **DTLS (Datagram Transport Layer Security)**.
- **Overlay Management Protocol (OMP):** The proprietary routing protocol running over the control connections (TLS/DTLS tunnels) used to exchange routing information between edges and vSmart. "Why overlay management? Because at the end of the day we are creating overlays."
- **Data Tunnels:** The tunnels carrying actual user data traffic, typically built using **IPsec** between edge routers.
- **Fabric:** The term "fabric" refers specifically to the **overlay data connections** (yellow data tunnels) creating the full-mesh or manipulated topology. "Fabric is just nothing whatever the overlay you are creating is your fabric not the control connection the data connection part."
- **vBond Orchestrator:** While not detailed in this excerpt, the vBond is mentioned as a crucial element responsible for the automated flow, ensuring that edge routers know where to make control connections with vSmart and vManage. This is sometimes referred to as the **orchestration plane**.

7. Key Takeaways and Conclusion

- SD-WAN simplifies WAN deployment and management through centralization and automation.
- Understanding the distinction between **control plane** (intelligence, routing table creation) and **data plane** (packet forwarding) is fundamental.
- The ability to **manipulate next hops** from a central controller is key to dynamically shaping network topologies (e.g., hub-and-spoke from a full-mesh default).
- **Tunneling** creates secure "overlay" networks over the "underlay" internet, providing logical direct connectivity for both control and data traffic.
- Cisco SD-WAN (Viptela) utilizes **vSmart** as the control plane, **vManage** as the management plane (single pane of glass), and **vEdge/cEdge** routers as the data plane elements.
- **OMP** is the proprietary routing protocol for control plane information exchange, and **IPsec** is used for data plane tunnels.
- The entire overlay network created by the data tunnels is known as the **SD-WAN fabric**.
- The concepts learned (control plane, data plane, next hop, tunneling) are largely vendor-agnostic and apply to many other software-defined networking technologies.

The instructor highly recommends re-watching the session and becoming comfortable with the "board" illustrating the combined SD-WAN solution, as it forms the bedrock for understanding future, more advanced topics.

Day 5 & 6:

SD-WAN Architecture and Zero-Touch Provisioning (ZTP) with a Focus on Security Fundamentals

Executive Summary

This session provides a deep dive into the Cisco SD-WAN architecture, specifically focusing on the crucial role of Vbond as the orchestrator and Vmanage as the management plane, enabling Zero-Touch Provisioning (ZTP) for edge devices. A significant portion of the briefing is dedicated to understanding the fundamental concepts of network security—Confidentiality, Integrity, and Authentication (CIA)—which are the building blocks of secure communication within the SD-WAN fabric. The speaker emphasizes that understanding these security principles is essential for comprehending how SD-WAN ensures secure connectivity and automated device onboarding.

Key Themes & Important Concepts

1. Cisco SD-WAN Architecture Components

The Cisco SD-WAN solution consists of three primary controller components and edge devices:

- **Vbond (Orchestration Plane):**
 - Acts as the "orchestrator," arranging and combining all pieces of the SD-WAN solution to achieve a desired, often automated, result.
 - Its primary function is to provide edge devices (Vedges/Cedges) with the IP addresses of the Vmanage and Vsmart controllers.
 - "The Vbond is on only there to tell you the information about V manage and V smart and that is it."
 - The connection to Vbond is not permanent; once the edge device receives the necessary controller information, the Vbond's responsibility is largely complete for that device.
 - In a fully automated (Zero-Touch Provisioning) scenario, the Vbond address itself is discovered through an internet-based "Vbond of Vbonds" (e.g., [viptella.com / helperviptella.com](http://viptella.com/helperviptella.com)).
- **Vsmart (Control Plane):**
 - Responsible for exchanging routes and controlling the SD-WAN fabric.
 - Edge devices create TLS/DTLS control tunnels with the Vsmart to exchange routing information using the OMP (Overlay Management Protocol).
 - "This guy knows everything. This guy knows all the routes everything of your complete van infrastructure right and it is going to control your complete van infrastructure."
- **Vmanage (Management Plane):**
 - Provides a single pane of glass User Interface (UI) for managing, configuring, troubleshooting, and monitoring the entire SD-WAN network.
 - It is capable of configuring all other controllers (Vsmart, Vbond) and edge devices.
 - "This device is capable of configuring anything on this V smart vond and on these devices itself... You can monitor your complete van infrastructure. You can create designs for example what exactly you need? You need hub and spoke or you need full mesh or you need custom topology."
 - Vmanage pushes configuration templates (e.g., LAN-side IP addresses, routing protocols like OSPF) to edge devices using the Netconf protocol.
- **Vedge/Cedge (Data Plane):**
 - These are the edge devices (Vedges are Viptella devices, Cedges are Cisco devices) that establish tunnels to the controllers and carry actual data traffic.
 - They form TLS/DTLS control tunnels to Vsmart and IPsec data tunnels for traffic flow.
 - "These are your data tunnels... The data tunnels are showcasing uh uh in this diagram uh showcased in diagram by this yellow."

2. Zero-Touch Provisioning (ZTP) / Plug and Play (PnP)

- **Goal:** To fully automate the onboarding of new edge devices without manual configuration.
- 1. **Process Overview:** An unconfigured edge device is powered on and obtains an IP address, default gateway, and DNS information via DHCP from its local network (e.g., service provider, home router).
- 2. The device (hardcoded to do so) automatically connects to a "Vbond of Vbonds" server on the internet (e.g., viptella.com / helper.viptella.com) via a secure tunnel (programmed into the device's image).
- 3. This "Vbond of Vbonds" server contains a database mapping device serial numbers to company names and their respective Vbond IP addresses.
- 4. The edge device sends its serial number to the "Vbond of Vbonds," which then replies with the IP address of the organization's specific Vbond.
- 5. The edge device then establishes a secure connection with its assigned Vbond, sends its serial number, and in return receives the IP addresses of its Vmanage and Vsmart controllers.
- 6. The edge device then forms control connections (TLS/DTLS tunnels) with the Vmanage and Vsmart.
- 7. Vmanage, recognizing the device by its serial number, pushes pre-configured templates (e.g., LAN-side configuration, routing protocols) to the device via Netconf, making it a fully operational part of the SD-WAN network.
- **Benefits:** Reduces deployment time and eliminates the need for on-site configuration, enabling rapid scaling of branches. "Many sites were bring up in less than 10 minutes. We just need to power them on."
- **On-Premises ZTP:** For organizations with strict security requirements (e.g., banking), controllers (Vmanage, Vsmart, Vbond) can be deployed on-premises. To maintain ZTP functionality in an on-premises environment, a local "Vbond of Vbonds" server (like helper.viptella.com) would also need to be deployed and managed by the organization.

3. Foundational Network Security Concepts (CIA Triad)

The speaker emphasizes the critical role of security in SD-WAN to prevent unauthorized devices from joining the network and accessing sensitive data. The core building blocks of network security are:

- **Confidentiality:**
 - Ensures that only the intended sender and receiver can read a message.
 - Achieved through **Encryption**, the process of transforming plain text into an unreadable format (ciphertext).
 - **Algorithm:** The specific process used for encryption (e.g., moving letters left).
 - **Key:** A secret value that determines how the algorithm is applied (e.g., how many positions to move left/right). Without the correct key, the encrypted message cannot be decrypted.
 - Analogy: German Enigma code in WWII.
- **Integrity:**
 - Ensures that a message has not been altered or tampered with during transmission.
 - Achieved through **Hashing**, a one-way mathematical function that produces a fixed-size "hash value" (or "fingerprint") for any input message.
 - "If I change any number into the original message this output is going to be totally totally different."
 - The sender calculates a hash of the original message and sends both the encrypted message and the hash. The receiver decrypts the message, calculates its own hash, and compares it to the received hash. If they don't match, the message's integrity has been compromised.
- **Authentication:**
 - Ensures that the communicating parties are genuinely who they claim to be.
 - Addresses the "problem of trust"—how to verify the identity of the other party in a communication.
 - Analogy: A village headman (trusted third party) vouching for a businessman and a supplier.
 - In computer networks, this role is played by a **Certificate Authority (CA)**, a trusted entity that issues digital certificates to verify identities.

4. TLS and DTLS Tunnels in SD-WAN

- **TLS (Transport Layer Security):** A secure protocol that typically runs over TCP. Used for control connections between Vmanage/Vsmart and edge devices.
- **DTLS (Datagram Transport Layer Security):** A secure protocol designed to run over UDP. Used for the initial connection between edge devices and the Vbond due to UDP's connectionless nature, which can be more efficient for initial discovery in some scenarios.
- Both TLS and DTLS provide Confidentiality, Integrity, and Authentication.

Day 7 & 8:

Network Security Fundamentals for SD-WAN

This briefing summarizes key concepts in network security, including symmetric and asymmetric encryption, Public Key Infrastructure (PKI), digital certificates, hashing, and their practical application in Transport Layer Security (TLS/DTLS) and IPSec for SD-WAN solutions. The ultimate goal is to understand how trust, confidentiality, and integrity are established in network communications, particularly within the context of bringing up SD-WAN controllers (vSmart, vManage, vBond, and edge devices).

1. Core Security Concepts: Confidentiality, Integrity, and Authentication

The foundational elements of network security are:

- **Confidentiality:** Ensuring that data remains private and is not accessible to unauthorized parties.
- **Integrity:** Guaranteeing that data has not been tampered with or altered during transit.
- **Authentication:** Verifying the identity of the communicating parties to establish trust.

The source emphasizes that these concepts are often confused, especially when public and private keys are introduced. A significant portion of the discussion is dedicated to clarifying the distinction between confidentiality and authentication.

2. Encryption Types: Symmetric vs. Asymmetric

2.1. Symmetric Encryption:

- **Definition:** Uses a single, identical key for both encryption and decryption.
- **Mechanism:** "Rahul wants to send a message to Anjel... he is applying a key and an algorithm so that this message will become M-dash... An Anjeli can is going to pass this message... to the same key value and to the same algorithm."
- **Characteristics:Fast:** "this symmetric encryption is pretty fast. Why fast? Because you are having a single key at both ends."
- **Key Distribution Challenge:** The primary problem is securely sharing the single key between communicating parties.

2.2. Asymmetric Encryption (Public Key Cryptography):

- **Definition:** Uses two mathematically linked keys: a public key and a private key.
- **Mechanism:** "you are going to have two different keys. One key to encrypt the message and the and and and the other key to decrypt the message."
- **Key Properties:Public Key (e.g., X):** "Public key can be distributed to everybody because this should be publicly available."
- **Private Key (e.g., Y):** "the private key can be used with this person only right it should not be shared because it's a private key."
- **Interrelation:** The keys (X and Y) are "interrelated numbers... if I encrypt a message with a particular number, it is only going to be decrypted with the corresponding pair... if I encrypt you using X it is going to be decrypt it is going to be only decrypt using Y not even X... and if I encrypt any number with Y it is going to be decrypt only with X nothing else."
- **Computational Cost:** "asymmetric key encryption or decryption is very very much computationally strong."

3. Public Key Infrastructure (PKI) and Digital Certificates: Building Trust

3.1. The Trust Problem:

- In computing systems, establishing trust between unknown parties (e.g., Rahul and an HDFC bank server) is crucial. Unlike human interactions where a "village headman" might vouch for someone, mathematics is used to build trust in digital environments.

3.2. Certificate Authority (CA): The "Village Headman" of Trust:

- **Role:** A trusted third party that verifies identities and issues digital certificates. "this is as good as that old man who is the village headman who was building the trust between Anjali and Rahul."
- 1. **Process of Certification (simplified):**An entity (e.g., HDFC bank server) approaches the CA with its public key (K) and identity information (name, location, website).
- 2. The CA verifies the entity's identity (e.g., by checking company papers).
- 3. Upon successful verification, the CA "signs" the entity's public key and associated information. This signing process involves encrypting the *hash* of the entity's public key (and other data) using the CA's *private key* (N). "The certificate authority is going to sign this public key of TFC bank server with its private key."
- 4. The result is a **digital certificate (K-dash)**: a file containing the entity's public key and a digital signature from the CA.

3.3. Certificate Validation (Authentication):

- **Mechanism:** When a client (e.g., Rahul) wants to communicate with a server (e.g., HDFC bank), the server sends its digital certificate.
- **Client's Role:** Rahul possesses the CA's *public key* (M), typically pre-installed as a "root certificate" in web browsers and operating systems. "certificate authorities public key which is M is publicly available... already available here with Rahul."
- Rahul uses the CA's public key (M) to decrypt the digital signature (H-dash) on the certificate. If successful, this proves the certificate was signed by the CA's private key (N). "If I am able to decrypt it using M, it was encrypted using N."
- **Trust Establishment:** Since Rahul trusts the CA, and the CA's signature is validated, Rahul can trust that the public key (K) within the certificate genuinely belongs to the HDFC bank server. "This message is of no other value than authenticating HDFC bank that you are talking to SDFC bank."

3.4. Digital Signature and Hashing:

- **Hashing:** A one-way function that takes an input (e.g., a file or public key) and produces a fixed-size, unique "fingerprint" called a hash value (H). "Instead of encrypting this complete file, what MR certificate authority is doing, it is taking out the hash from this file and that hash is a little bit little number."
- **Digital Signature:** The hash value is then encrypted using the signer's (CA's) private key. This encrypted hash is the digital signature.
- **Integrity (Indirectly):** If any part of the original data or certificate is altered, recalculating its hash will yield a different result, invalidating the signature and indicating tampering. This is crucial for verifying integrity.

4. Confidentiality vs. Authentication Revisited

- **Authentication (Private Key for Encryption):** When signing a certificate, the CA uses its *private key* to encrypt the hash. This proves the CA's identity because only its corresponding public key can decrypt it. The signed data (the public key of the server) is *not confidential* because it's meant to be publicly available. "I am not providing confidentiality here confidentiality means encryption... I am providing authentication here which is a trust."
- **Confidentiality (Public Key for Encryption):** To achieve confidentiality, the sender encrypts the message with the *receiver's public key*. Only the receiver, who possesses the corresponding *private key*, can decrypt the message. "Mr. Rahul is encrypting this message with HTFC bank server public key which is K... now it can only be decrypted using the private key of SDFC bank server."

5. Combining CIA for Secure Communication: TLS (Transport Layer Security)

TLS is a protocol suite that combines authentication, confidentiality, and integrity to provide secure communication over insecure networks like the internet.

5.1. TLS Handshake (Phases of Communication):

1. **TCP Three-Way Handshake:** Before TLS, a standard TCP connection is established (SYN, SYN-ACK, ACK). "TLS works on TCP. It means that TLS cannot be formed unless there is a TCP connection."
2. **Client Hello:** Rahul sends a "Client Hello" message, specifying its supported encryption algorithms (e.g., AES), integrity algorithms (e.g., SHA), authentication methods (e.g., RSA), and TLS version. These are known as **cipher suites**.
3. **Server Hello & Certificate:** HDFC bank replies with a "Server Hello," confirming supported cipher suites, and sends its digital certificate.
4. **Certificate Validation:** Rahul validates the HDFC certificate using the CA's public key (root certificate). This establishes **authentication**.
 - **Symmetric Key Exchange (Confidentiality Setup):** Rahul generates a **secret symmetric key (S)**.
 - Rahul encrypts this secret key (S) using HDFC bank's **public key (K)** (obtained from the certificate).
 - Rahul sends the encrypted secret key (S-dash) to HDFC bank.
 - HDFC bank decrypts S-dash using its **private key (L)**, recovering the secret key (S).
 - **Rationale:** Asymmetric encryption is computationally expensive. It's used *only* for the initial authentication and the secure exchange of a symmetric key. Subsequent communication then uses the faster symmetric encryption. "we are going to first use this asymmetric algorithm for the authentication purpose to share the keys and then basically we are going to use the symmetric key."
1. **Secure Data Transfer:** All subsequent messages between Rahul and HDFC bank are encrypted and decrypted using the shared symmetric secret key (S), providing **confidentiality**. Hashing (SHA) is used for **integrity** checks.

5.2. DTLS (Datagram Transport Layer Security):

- **Difference from TLS:** DTLS functions similarly to TLS but works over UDP instead of TCP.
- **No Connection Establishment:** "DTLS doesn't require a TLS connection at all... DTLS is designed to work on UDP and UDP basically you know there is no connection establishment happens at the first place."
- **SD-WAN Application:** "the initial connection towards the Vbond which makes which is made by this edges it's actually the DTLS connection."

6. IPSec (Internet Protocol Security): Data Tunnels

- **Function:** IPSec operates at the **IP layer (network layer)** and provides confidentiality, integrity, and authentication for data traffic. "Another technology which works at the IP layer and IP layer is by the way guys your network network layer right which works at the IP layer and provides the same functionality same confidentiality integrity and authentication."
- **SD-WAN Application:** In SD-WAN, IPSec tunnels are used for **data plane** communication between edge devices (vEdges). "this yellow link is purely these are purely for data transfer and what this tunnel is this tunnel is IP sec."
- **Key Exchange in SD-WAN:** Unlike traditional IPSec which might use IKE (Internet Key Exchange), SD-WAN leverages the pre-existing secure TLS/DTLS control connections to simplify key management. "this VS smart being a smart device is taking care of this key exchange problem right it is distributing the keys to all the vages so that they can communicate."
- **Tunnel Mode:** SD-WAN IPSec tunnels operate in tunnel mode, where "you are tunneling a packet inside another packet."
- **No MTU/MSS Impact:** Since IPSec adds headers at the IP layer, it does not typically affect TCP MSS calculations, which are handled by TCP itself. "this protocol is actually the application layer protocol... TCP is already taking care... it is not going to impact." (Note: The speaker initially incorrectly states IPSec is application layer then corrects to IP layer. The MTU/MSS impact discussion refers to application layer protocols generally, but for IPSec, the statement is that it won't impact TCP's MSS calculations, implying TCP adapts).

7. SD-WAN Lab Setup Considerations

The course prepares participants to build an SD-WAN lab, requiring specific resources:

- **Hardware Requirements:** Minimum 8-12 CPUs, 32 GB RAM, 120 GB HDD.
- **Cloud Option:** Google Cloud Platform (GCP) is recommended for those without sufficient local hardware, offering a free trial of up to \$1400 for 3 months.
- **Software:** Installation of EVE-NG (or similar virtualization platform) and SD-WAN images (vSmart, vManage, vEdge, cEdge).
- **Learning Curve:** Prior knowledge of cloud networking is beneficial for using GCP.

Conclusion

This session provides a comprehensive grounding in the cryptographic principles and protocols (TLS/DTLS, IPSec) that underpin secure communication in modern networks, particularly in the context of SD-WAN deployments. Understanding how authentication, confidentiality, and integrity are achieved through symmetric/asymmetric encryption, PKI, digital certificates, and hashing is fundamental before configuring and troubleshooting SD-WAN solutions. The emphasis on practical application and the combination of these concepts into real-world protocols like TLS and IPSec highlights their importance in establishing trusted and secure control and data plane connections.

Day 9 & 10:

Cisco SD-WAN Lab Setup on GCP

1. Introduction to the Lab and its Environment

The session, led by Vishno Dut, focuses on building a Cisco SD-WAN lab from scratch, primarily on Google Cloud Platform (GCP). The lab design emphasizes foundational concepts and aims for a basic setup that covers core SD-WAN functionalities without excessive resource demands.

- **Target Audience:** Individuals with some understanding of network security concepts and potentially prior experience with network simulators like EVE-NG.
- **Lab Focus:** Fundamentals of Cisco SD-WAN, including controller setup, edge device integration, and basic networking configurations.
- **GCP as the Platform:** GCP (Google Cloud Platform) is chosen for hosting the lab due to its accessibility and free tier benefits. The presenter highlights that concepts discussed for GCP are common across other cloud providers like AWS and Azure.

2. Core GCP Concepts for Lab Setup

Understanding basic cloud concepts is crucial for effectively setting up the lab.

- **Google Cloud Platform (GCP):** A public cloud service that provides infrastructure to run virtual machines and applications. Google manages the underlying physical servers, saving users the cost and responsibility of managing their own data centers (space, cooling, operational staff).
- **Infrastructure as a Service (IaaS):** Google provides the raw computing infrastructure (servers, storage, networking) as a service, allowing users to deploy and manage their virtual machines (VMs) on top of it.
- **VMs (Virtual Machines):** Applications and servers previously run on physical hardware in a data center are now run as VMs on the cloud hardware. These VMs are provisioned on powerful physical servers (e.g., 128 CPU, 512 GB RAM, 12 TB hard drive) managed by Google.
- **Hypervisor (ESXi):** A thin operating system layer installed on physical servers by Google (or any cloud provider) that enables running multiple virtual machines with different operating systems (Linux, Windows) on a single physical machine, efficiently utilizing hardware resources.
- **Regions and Availability Zones:** GCP's global infrastructure is divided into geographical regions, each containing multiple availability zones (data centers).
- **Regions:** Represent distinct geographical areas (e.g., Asia, Europe, Americas). Users should choose a region closest to them to minimize latency.
- **Availability Zones (AZs):** Corresponds to one or more data centers within a region. They provide redundancy; if one data center fails, the VM can run in another DC within the same AZ, as it's Google's responsibility to ensure VM availability.
- **Virtual Private Cloud (VPC):** A logically isolated network within GCP that allows users to define their own network topology, including VLANs, switches, routers, firewalls, and load balancers, similar to their on-premises data center.
- **"Whatever the network you have created inside this Google cloud, it is going to be your network or it is going to be your personal cloud and that is why we call them as a virtual private cloud."**
- Multiple VPCs can exist on the same public platform, each belonging to a different company (e.g., Netflix on AWS).
- When creating a VM, users must select a VPC.
- **Networking Components within VPC:**
 - Inbuilt Firewall:** Safeguards VMs from external attacks. Users can configure Ingress (inbound) and egress (outbound) rules. GCP firewalls are stateless, requiring explicit rules for both directions of traffic.
 - Internet Gateway Router:** Provided by Google, dedicated to the user's VPC, allowing VMs to communicate with the internet. Users can modify routing rules.
 - IP Addresses:** VMs are assigned both a private IP address (for internal VPC communication) and a public IP address (for internet access).
 - GCP Free Tier & Lab Resources:** Google offers a \$300 credit for 3 months to new users for exploration.
 - Providing a company email ID can add an extra 100, totaling 400 for 3 months.
 - The SD-WAN lab requires at least **12-16 CPUs, 32GB of RAM, and 128GB of hard disk space**. Users are advised to utilize the free tier if their local machine cannot meet these requirements.
 - Due to resource limitations for free-tier customers, users might need to try different availability zones or regions to find available hardware.

3. EVE-NG: The Lab Emulator

EVE-NG (Emulated Virtual Environment for Network, Security and IoT) is the chosen software for building the SD-WAN lab.

- **Purpose:** A simulator/emulator used to build and run Cisco devices (and other vendors).
- **Installation:** EVE-NG is installed on an Ubuntu virtual machine within GCP.
- **Nested Virtualization:** This is a critical step. Since EVE-NG runs virtual instances of network devices, and EVE-NG itself is running on a VM in GCP, "nested virtualization" must be enabled. This allows running a "virtual machine inside a virtual machine."
- **Accessing EVE-NG:** Once EVE-NG is installed, it can be accessed via its public IP address through a web UI (username: admin, password: eve).
- **Device Images:** Before building the lab topology, actual operating system images for SD-WAN components (vManage, vSmart, vBond, vEdge/cEdge, Linux host for UI access, routers, switches) must be transferred from the user's local machine to the EVE-NG VM.

4. SD-WAN Lab Components and Topology

The lab focuses on a fundamental SD-WAN topology:

- **Controllers:** vManage, vSmart, vBond. These are logically grouped as the "controller site."
- **Data Center (DC):** At least one DC site.
- **Branch Sites:** At least two branch sites.
- **Transports:** Two different WAN transports: MPLS and Internet.
- **Management Network:** A separate network for managing controllers and accessing the vManage UI.
- **Host Machine:** A Linux machine (with Chrome/Firefox) is needed within the lab to access the vManage web UI.
- **Edge Devices:** The lab will primarily use vEdges, but the process for cEdges (Cisco routers) is similar.
- **Scalability:** The current topology is designed to be resource-efficient (max 10 CPUs) but can be expanded with more branches or edges for exploring additional use cases.
- **ZTP (Zero Touch Provisioning):** The speaker mentions that while ZTP for full automation is possible on-premises (requiring a local ZTP server/DNS for helper.vhiptella.com resolution), it will not be the primary focus of this lab's initial setup to demonstrate manual configuration.

5. SD-WAN Controller Deployment Models and Responsibilities

The briefing clarifies the deployment models for SD-WAN controllers:

- **Cisco Cloud-Hosted (Recommended):** Cisco runs and manages the controllers in their private cloud. This offloads management and scaling responsibilities from the customer. Users are provided access to vManage.
- **"If you go with Cisco and if you need more VSmart if your network is growing if you need more Vbond more VManage it is going to be the responsibility of Cisco to scale it up."**
- **On-Premises Deployment (Lab Focus):** The customer (or in this case, the lab builder) hosts and manages all controllers (vManage, vSmart, vBond) on their own infrastructure. This gives full control but also full responsibility for management, scaling, and most critically, **certificate management**.
- **"We are going to do all these things because we are creating this lab and this is how you are going to learn things."**

6. Certificate Management and Trust in SD-WAN Fabric

A significant portion of the session is dedicated to the critical role of certificates in establishing trust within the SD-WAN overlay.

- **Importance of Trust:** SD-WAN controllers and edge devices need to mutually authenticate and trust each other to build the fabric. This is achieved through certificates.
 - **Certificate Authority (CA):** Acts as a trusted third party that issues and signs digital certificates.
1. **Options for CA in Lab/Linux Host:** A dedicated Linux machine can be configured as the CA (e.g., using XCA tool).
 2. **vManage (Chosen for Lab):** vManage itself can act as the CA, signing its own certificate (self-signed) and certificates for other devices. This is the "pretty straightforward method" chosen for the lab videos.
 3. **Certificate Process: Root Certificate Distribution:** The CA's root certificate (containing its public key) must be installed on all SD-WAN devices (vManage, vSmart, vBond, vEdges) that need to establish trust.
 4. **Device Certificate Generation:** Each device generates a Certificate Signing Request (CSR) containing its public key and sends it to the CA.
 5. **CA Signing:** The CA signs the CSR, issuing a device-specific certificate.
 6. **Mutual Authentication:** When devices communicate (e.g., vSmart to vBond), they present their device certificates. The receiving device validates this certificate using the installed root CA certificate, ensuring the identity and trustworthiness of the communicating party.
- **Necessity for Network Engineers:** The presenter stresses that understanding certificate concepts (public key, private key, hashing, integrity, TLS/DTLS) is fundamental for any network engineer, especially with the prevalence of security in modern architectures.
 - "Most of the things most of the trainings even even the training which happens at Cisco on these things people do not talk about all these things right they give you the lab directly but literally if you are a network engineer you should understand it..."

7. Virtual Routing and Forwarding (VRF) and SD-WAN VPNs

VRF is a fundamental concept for segmenting networks on a single physical router, crucial for understanding SD-WAN VPNs.

- **Traditional Router without VRF:** A single routing table (Global Routing Table) for all interfaces.
 - **Need for VRFs:** To separate different networks/customers/traffic types (e.g., Enterprise vs. Guest, IoT), effectively turning one physical router into multiple virtual routers, each with its own independent routing table.
- "You want that your guest network or the traffic of the guest network it is always separate than the traffic of your enterprise network."
 - **VLAN Analogy:** Similar to how VLANs segment a single physical switch into multiple virtual switches at Layer 2, VRFs segment a single physical router into multiple virtual routers at Layer 3.
 - **VRF-Lite:** A manual configuration approach where VRFs must be explicitly defined on every router in the path.
 - **MPLS L3VPN:** A more scalable solution for service providers where VRFs are defined only on Provider Edge (PE) routers, with intermediate Provider (P) routers not needing VRF awareness due to MPLS labels.
 - **SD-WAN VPNs:** SD-WAN routers (vEdges/cEdges) inherently utilize VRFs, referred to as VPNs in SD-WAN terminology.
 - **Default VPN: VPN 0:** The "transport side" VPN where WAN interfaces (Internet, MPLS) connect. All WAN-bound traffic flows through VPN 0.
 - **Management VPN (VPN 512):** A reserved VPN for out-of-band management of the SD-WAN device itself (e.g., connecting to vManage).
 - **Service VPNs:** User-defined VPNs (e.g., VPN 1, VPN 2, VPN 3) on the "LAN side" of the router. These are used to segregate different customer networks or traffic types (e.g., enterprise, guest, IoT) within the local area network.
 - "This router by default is divided into two parts it is not the case that it is only your GRT is there right VPN zero and management VPN..."
 - The ability to send traffic from multiple LAN-side Service VPNs over a single WAN-side VPN 0 link is a core SD-WAN capability.

8. Key Takeaways and Call to Action

- **Hands-on Practice:** The presenter strongly encourages participants to build the lab themselves, either on their own hardware or using GCP's free tier.
- "It is my expectation that please create this lab. Doesn't matter wherever you are creating it. It it really doesn't matter but yes do create it because now whatever the concept we are going to see mostly we are going to see them on on on this VManage VS and VSmart."
- **Learning from Mistakes:** Making mistakes during the lab setup is encouraged as it fosters deeper understanding.
- "When you do mistake and when you try to think that why it happens why it is not working then most of the time you learn."
- **Course Structure:** The course will cover fundamentals (18-20 hours) and then delve into policies and design (last 10 hours), potentially extending beyond the initial 24 hours. Future videos for SD-WAN features or integrations will be free for existing course participants.
- **Importance of Fundamentals:** The concepts covered (cloud basics, EVE-NG, certificates, VRFs) are foundational building blocks for understanding the rest of the SD-WAN solution and broader networking architectures.
- **Multi-Architecture Skills:** Growing in one's career often involves gaining horizontal knowledge across different domains (data center, wireless, security, enterprise, service provider) to combine technologies for complete solutions.

Day 11 & 12:

SD-WAN Lab Setup and Core Concepts:

This briefing summarizes key aspects of setting up an SD-WAN lab, focusing on initial configurations, the role of VManage, certificate management, and fundamental architectural components. It emphasizes practical steps, troubleshooting, and the importance of foundational knowledge in SD-WAN.

1. The SD-WAN Lab Environment and Initial Setup

The session primarily focuses on practical, hands-on configuration to bring up an SD-WAN lab environment. While SD-WAN solutions often boast "zero-touch provisioning" (ZTP), setting up the core components (VManage, VSmart, VBond) in a lab requires initial manual configuration.

- **Lab Components:** The typical lab topology includes:
 - **VManage:** The central management console for the SD-WAN fabric.
 - **VSmart:** The control plane orchestrator.
 - **VBond:** The orchestrator that helps devices discover VManage and VSmart.
 - **vEdge/cEdge (VH/CH):** The edge devices that form the SD-WAN fabric.
- **Linux Host:** A desktop machine deployed within the lab environment, primarily for accessing the VManage UI via a web browser (e.g., Firefox/Chrome).
- **Network Infrastructure:** Switches and routers forming the underlay network, connecting these components.
- **Initial Configuration Philosophy:** The instructor emphasizes understanding "what is behind those commands" rather than just providing commands. This foundational understanding is crucial for effective troubleshooting and design. The goal for the initial lab setup is to make it "up and running," which alone constitutes "80% of the solution."
- **Linux Host Setup:** A Linux desktop machine is essential for accessing the VManage GUI. It requires:
 - Configuring an IP address on its network interface (e.g., ENS3) within the management network range. This involves editing a configuration file in `/etc/netplan/` and applying changes using `sudo netplan apply`.
 - Default login credentials for the Linux machine are "user" (username) and "Test123" (password).
- **File Transfer:** The lab setup facilitates easy file transfer (e.g., serial files containing vEdge lists) from the user's local machine to the Linux host within the lab via RDP, by enabling shared drives.

2. VManage: The Single Pane of Glass

VManage serves as the central point for managing and monitoring the entire SD-WAN fabric.

- **Accessing VManage:** VManage is typically accessed via a web browser (Firefox/Chrome) using an HTTPS request to its management IP address and port (e.g., `https://192.168.1.11:8443`).
- **Management Network (VPN 512):** VManage has a dedicated management interface (e.g., Ethernet0) configured as part of VPN 512. This separation of management and production traffic is a common industry standard to ensure network manageability even if the production network is overloaded. Initial configuration involves assigning an IP address and optionally a default route for external access.
- **Production/Transport Network (VPN 0):** VManage also has an interface (e.g., Ethernet1) configured as part of VPN 0, which is the default transport VPN where overlay tunnels are formed. Initial configuration includes an IP address, and critically, enabling "tunnel" and "allow services all" on this interface, though it's noted that "allow all" might not be ideal in production and specific services might need explicit mention even with "allow all."
- **Dashboard Overview:** The VManage dashboard provides a high-level view of the SD-WAN fabric, including:
 - Number of VManage instances (can be clustered).
- **Control Connections:** Status of TLS/DTLS connections between controllers and between edges and controllers (Up, Partial, Down).
- **IPSec Connections:** Status of data plane connections between devices (Full, Partial, No connectivity). IPSec is where "your data is flowing."
- **WAN Circuit Utilization:** Bandwidth usage, latency, loss, and jitter – critical parameters for application-aware routing and traffic steering.
- **Configuration and Monitoring:** VManage is used for:
 - **Device Management:** Uploading vEdge serial numbers/chassis IDs, adding controllers (VSmart, VBond).
 - **Templates & Policies:** The primary mechanisms for configuring devices and defining network behavior. Policies are crucial for actual network design and tweaking the default "full mesh behavior" to meet customer requirements. The instructor states, "remaining 60% is going to be your policies."
 - **Monitoring Section:** Viewing device locations on a map, events, alarms, and audit logs.
 - **Troubleshooting Tools:** VManage offers integrated troubleshooting capabilities.
 - **SSH Terminal:** Direct SSH access to devices from the UI.
 - **Operational Commands:** GUI-based access to show commands for underlay and overlay networks.
- The instructor stresses that effective troubleshooting relies on "solid fundamental knowledge" and understanding the problem statement, rather than just comparing configurations. Debugging tools (like debug OSPF in a traditional context) are highlighted as essential.

3. Core Architectural Components and Their Roles

Understanding the distinct roles of VManage, VSmart, and VBond is fundamental.

- **VManage:** The **central management plane**, acting as the "single piece where you are going to log into" for all configuration, monitoring, and troubleshooting. It pushes commands to VSmart.
- **VSmart:** The **control plane orchestrator**. It "knows your fabric. It knows everything in your fabric." VManage pushes desired topology and policy configurations to VSmart, which then implements them across the fabric.
- **VBond:** The **orchestrator/zerotouch provisioning server**. Its role is "pretty limited" and "just to orchestrate things, just to bind this solution." It acts as the initial point of contact for vEdge devices, providing them with the IP addresses of VManage and VSmart so they can establish further connections. The connection between vEdges and VBond is "not permanent" and is torn down after initial discovery.

4. Initial Configurations: System Parameters and NTP

Beyond interface configurations, several system-wide parameters are crucial for all SD-WAN components (VManage, VSmart, VBond, vEdge).

- **System Parameters:**
 - **Hostname:** A descriptive name for the device.
 - **System IP:** A unique, loopback-like IP address for identifying the device within the SD-WAN fabric. Similar to router-id in traditional routing protocols like OSPF/BGP, it doesn't need to be routable but uniquely identifies the device.
 - **Site ID:** A numerical identifier for a specific network location (e.g., a data center, a branch). This helps in organizing and applying policies across different sites. "This is going to be very very important. When we talk about policies."
 - **Organization Name:** The name of the SD-WAN overlay network.
 - **Clock Time Zone:** Essential for time synchronization.
 - **VBond Address:** All controllers and vEdges need to know the VBond's IP address for initial discovery and communication.

- **local keyword (VBond specific):** When configuring VBond, specifying local along with its VBond address indicates that this device is the VBond for the infrastructure.
- **NTP (Network Time Protocol):**
 - NTP is **critical** for time synchronization across all SD-WAN devices. Certificates have validity periods based on timestamps, and inconsistent time can lead to certificate validation failures, breaking the system.
 - "Timing is important, NTP is important. If you do not do that then it might be problem with your certificate."
 - The lab setup typically involves configuring a router in the underlay as an NTP server, and then configuring all SD-WAN components to synchronize their time with it.

5. Certificates: The Backbone of SD-WAN Security

Security in SD-WAN is inherently tied to certificates, which are mandatory for device authentication and secure communication.

- **VManage as Certificate Authority (CA):** In a lab or on-premise deployment, VManage can act as its own Certificate Authority.
- **Key Generation:** The first step is to generate root CA keys (public and private keys) for the VManage using openssl (e.g., openssl genrsa -out root-ca.key 2048). These keys form the basis of the VManage's CA functionality.
- **Self-Signed Root CA Certificate:** A self-signed certificate for the VManage CA itself is generated using its own root CA keys. This "root CA needs to be installed everywhere" (on vEdges, VSmart, VBond) to enable them to authenticate VManage.
- **Certificate Signing Request (CSR):** Devices like VBond, VSmart, and VManage (as a device, not just the CA) need their own certificates to prove their identity.
 - These devices generate a Certificate Signing Request (CSR), which contains their public key, and send it to the VManage CA.
 - The VManage CA then signs the CSR using its private key, generating a valid certificate for that device.
- **Automated vs. Manual Process:** While Cisco's cloud-based SD-WAN deployments automate this process, in a lab or on-premise scenario, generating and signing certificates involves manual steps using openssl commands on the VManage console.
- **Single Pane for Certificate Management:** Once VBond and VSmart are added to VManage (via their IP addresses in the VManage UI), the VManage GUI allows generating CSRs for them with a "right-click and generate CSR" action. The signing process, however, still requires command-line interaction on VManage.
- **Importance of Security Knowledge:** The instructor strongly advocates for understanding certificate concepts, stating, "you cannot escape from this," and that this "security knowledge is going to be very very important for you."

6. Universal IOS-XE Image for Cisco Routers

Cisco has transitioned to a universal IOS-XE image for many of its routers (e.g., CSR, ASR).

- **Single Image, Dual Capability:** This single image contains code for both traditional IOS-XE and SD-WAN functionality.
- **Boot Mode Selection:** A router boots into standard IOS-XE by default. To enable SD-WAN, a command needs to be executed to set the device into "controller mode," upon which it will boot in SD-WAN mode.
- This simplifies image management for customers, as they don't need separate images for different deployment types.

7. Learning Philosophy and Expectations

The instructor outlines a specific learning approach and sets expectations for the course.

- **Hands-on First:** The immediate goal is to get the lab "up and running" and "make your lab up and running then you understand 40% of the solution."
- **Fundamentals are Key:** "If you understand the fundamentals well the remaining 70 kilometers are going to be easy for you." Without strong fundamentals, troubleshooting becomes "just comparing" rather than true problem-solving.
- **Course Scope:** The course aims to provide a "solid 30 kilometers" of SD-WAN knowledge, covering fundamentals, network design aspects, and migration. The remaining 70% requires individual deep dive and continuous learning, especially in policies.
- **Policies as Design:** Policies are highlighted as the "actual design" component of SD-WAN, allowing customization of traffic flow and network behavior beyond the default full-mesh.
- **Continuous Learning:** SD-WAN is an "evolving technology," and expertise requires ongoing engagement. Analogy of BGP knowledge (even after years, only 10% might be understood) is used to emphasize the depth of networking technologies.
- **Support and Flexibility:** The instructor is committed to extending the course duration if needed to ensure participants learn effectively, inviting questions and active participation. Documentation (text files, videos) is provided to support hands-on lab work.

Day 13 & 14:

SD-WAN Fundamentals: Templates, T-LOCs, and OMP

This briefing covers foundational SD-WAN concepts crucial for understanding network configuration and operation, focusing on the practical implementation of templates, the identification mechanism of Transport Locators (T-LOCs), and the core routing protocol, Overlay Management Protocol (OMP).

1. The Power of Templates in SD-WAN

Templates are a cornerstone of efficient SD-WAN deployment, designed to simplify and standardize network device configuration.

1.1. Addressing Legacy WAN Challenges: Traditional WAN environments face significant challenges, especially when scaling. Manually configuring each router for new branches is time-consuming, prone to errors, and requires onsite technical expertise.

- "Suppose this organization is a bank... they want to open 100 more branches... you need to plan how you are going to because these 100 branches are not close to me these are pretty remote."
- "Most of the times you forget something on this device and if you forget something on this device... you call the and who are the employees who are working on this remote branch? Those are bank employees who might who may know banking well but they do not know networking."

1.2. The Concept of Variables and Reusability: Many configurations across different devices share common parameters, with only specific values changing (e.g., IP addresses, neighbor IPs). These changing values are "variables."

- "The things which varies Even these interface doesn't vary right the things which vary is only this IP address right and the neighbor IP address... the things which varies along different configurations we call them variables."
- Templates allow for reusable configuration blocks, eliminating the need to write commands repeatedly for each device. "Can I do them once and reuse them again and again in different document? Of course, I can do."

1.3. Feature Templates vs. Device Templates: SD-WAN uses a hierarchical template structure:

- Feature Templates:** Small, reusable configuration blocks dedicated to a specific network feature (e.g., interface configuration, BGP/OSPF settings, banner messages).
 - "This template concept is not new. We have been doing this in the past but yes how Cisco's ST1 is implemented it it's really beautiful."
 - "This is dedicated to a particular feature. Which feature you are configuring? You are configuring an interface. Here this is this is again a template. Which template? This is a BGP feature template."
- Device Templates:** A comprehensive configuration for an entire device, built by calling multiple feature templates.
 - "This complete document is known as device template because this is the final thing which is going to go on your device."
- Variables within feature templates are populated when the device template is applied, often drawing data from external sources like Excel sheets. This allows for centralized management of unique device parameters.
- "You can have an excel sheet which says just write down IP address at fa0/interfaced and then your neighbor address neighbor one neighbor two whatever right and then you fill this document."

1.4. Zero-Touch Provisioning (ZTP): Templates enable ZTP, significantly simplifying device deployment. A new SD-WAN edge device (vEdge) can be shipped directly to a remote site, powered on and automatically receive its full configuration from vManage.

- "SD-WAN solution can. Most of the SD-WAN solution can. So whatever the device you buy from Cisco, you are just shipping it to the remote village where your branch is. You just powered it on."
- The vEdge, upon booting, obtains an IP address (often via DHCP) and communicates with a centralized ZTP server (e.g., at viptela.com) using its serial number and certificates. This allows it to discover and connect securely to its assigned vBond, vSmart, and vManage controllers.
- "The developer of this VH have hardcoded this thing that whenever you come up just go to viptela.com and there we have placed a server which we call as the ZTP server."
- "You just need the device to see the internet."
- Once connected, vManage pushes the pre-configured device template via NETCONF protocol over a secure control connection (DTLS/TLS).
- "How vManage is going to deliver this document to VH over this netconf protocol which is pretty secure over this green connections."
- This process eliminates the need for manual onsite configuration, reducing errors and deployment time.

2. Transport Locators (T-LOCs)

T-LOCs are a fundamental SD-WAN concept for uniquely identifying WAN transport connections, especially in dynamic environments.

2.1. The Problem with Dynamic IP Addresses: Traditional network identification relies heavily on static IP addresses. However, in modern WANs, particularly with internet links, IP addresses can be dynamically assigned via DHCP and may change upon reboots or lease expirations. This dynamic nature makes stable identification problematic.

- "If you talk about internet we are not sure that if we are getting the IP address from DHCP. This IP address can change."
- "If this IP address can change this should not be the criteria to identify the interface."

2.2. T-LOC Components for Unique Identification: To overcome this, SD-WAN introduces T-LOCs, a combination of three parameters that uniquely identify a specific transport circuit on an edge device:

- System IP:** A unique, static IP address assigned to each vEdge router, serving as its router ID within the SD-WAN fabric.
 - "System IP is just nothing but your router ID... System ID IP which uniquely identify the device. It should not be repeatable."
 - "If I say this edge one has system IP of this one, it is going to be unique across your SD-WAN fabric."
- Color:** A user-defined attribute (e.g., "biz-internet," "MPLS," "LTE," "gold," "silver") that categorizes or logically identifies the transport circuit. It helps distinguish different WAN connections on the same vEdge.
 - "This color is green. And this color is sky blue."
 - "With these two things you can identify a particular circuit IP address."
- Encapsulation:** The tunnel encapsulation type used for data plane connections, typically IPsec by default, but can also be GRE.
 - "We are also including the encapsulation here. Maybe the encapsulation type is maybe the IPsec."

2.3. T-LOCs and Dynamic IP Addresses: Even if the underlying interface IP address changes (e.g., due to DHCP renewal or reboot), the T-LOC (System IP + Color + Encapsulation) remains constant. This allows the SD-WAN fabric to consistently identify and establish tunnels to that transport circuit, automatically adapting to the new IP.

- "You are not dependent now on this IP address but still you can figure out what is the IP address configured at that particular time because you are not dependent on this variable."
- "T-LOC is just nothing or transport locator is just nothing. They are actually the they are pointing to actual IP address configured on that interface."

3. Overlay Management Protocol (OMP)

OMP is the proprietary routing protocol of Cisco SD-WAN, responsible for exchanging routing and topology information within the overlay network.

3.1. Centralized Control Plane: Unlike traditional distributed routing protocols (e.g., OSPF, BGP) that run directly between peer routers, OMP operates between the vEdges and the centralized vSmart controllers.

- "Instead of vmanage I have changed it to vsmart now we have smart... which knows every route in your um in your uh uh fabric or in your SD-wan network."
- "VS smart and VH are peering with each other. They are neighbors of each other. And which routing protocol they are running? They are running overlay management protocol."

3.2. Automatic Neighbor Adjacency and Route Exchange: Once a vEdge establishes secure control connections (DTLS/TLS tunnels) with vSmart controllers, OMP neighbor adjacencies are automatically formed. No manual configuration is required on the vEdges or vSmart for OMP peering.

- "The system The solution does it on your behalf."
- "VH is going to create a neighborship with VS smart. You do not need to configure anything for that even you do not need to configure anything Vmanage for that. It is going to be automated process."

3.3. Information Exchanged via OMP: OMP carries various types of information critical for overlay routing and tunnel establishment:

- **Virtual Routes (vRoutes):** These are the prefixes (LAN routes, static routes, OSPF intra-area routes) learned by the vEdges that need to be advertised across the SD-WAN fabric. By default, OMP advertises connected, static, and OSPF intra-area routes from the LAN side of the vEdges.
 - "What OM is doing... Whatever route you have here [LAN side]... it is going to be learned via this by this var is going to say that it has learned this route."
 - "By default, if you have not advertising this connected route also here or static route also here, nothing is going to be advertised into BGP... But here nothing. So what is the thought process behind it. The thought process behind it that once your system once your solutions comes up at least these LAN guys... should able to communicate with other VHS."
 - "We call these routes as virtual routes also."
- **T-LOCs:** As discussed, T-LOCs (System IP, Color, Encapsulation) identify the WAN transport interfaces on the vEdges. OMP advertises these T-LOCs along with their corresponding physical IP addresses to the vSmart controllers.
 - "It advertises T-LOCs and it advertise them separately."
- **Service Routes:** Information about services located behind vEdges, used for service chaining (not deeply covered in this segment).

3.4. vSmart as a Route Reflector/Orchestrator: The vSmart acts as a centralized route reflector, receiving all vRoutes and T-LOCs from connected vEdges. It then distributes this information to other vEdges based on configured policies, enabling inter-branch communication and determining tunnel destinations.

- "What VS smart is going to do whatever VS smart does it is going to advertise it so that this VH3 VH2 also learns about where exactly XXX is."
- "If you are aware of the concept of route reflector in BGP it is doing same."

3.5. Dynamic IPsec Tunnel Establishment: When a vEdge learns a vRoute for a remote destination via OMP (e.g., XXX.X/24 reachable via T-LOC A), it uses the T-LOC information to identify the destination vEdge's WAN interface IP address. It then dynamically establishes an IPsec tunnel to that IP address for data plane traffic.

- "When VH2 sees that XXX is reachable via T1 which is a TLO and the IP address corresponding to the TLO is 1.1.1.1 it immediately creates a tunnel with it."
- The dynamic nature of T-LOCs ensures that even if the underlying IP changes (e.g., due to DHCP), the tunnel can be re-established to the correct new IP address because the T-LOC itself remains consistent.

3.6. Policy Enforcement: vManage is used to configure routing policies. These policies are then pushed to the vSmart controllers, which apply them to the OMP routes they advertise. This allows administrators to control traffic flow, connectivity, and topology (e.g., full-mesh, hub-and-spoke) within the SD-WAN fabric.

- "Vmanage need to instruct this V smart that you need to perform such activities because my customer wants it."
- "If you tweak only these two things [virtual routes and T-LOCs], if you control VS smart, if you Specify VS smart that VS smart advertise only these T logs because on the basis of these T logs your IP tunnels are formed. If you control them you can control the topology on the basis of virtual routes you can control what devices should reach to what devices."

Day 15 & 16:

SD-WAN (OM) and Traffic Engineering:

This briefing document summarizes the key concepts and important facts regarding SD-WAN's Overlay Management Protocol (OM) and its application in traffic engineering, drawing directly from the provided audio transcript.

1. Introduction to OM and its Central Role

The Overlay Management Protocol (OM) is described as the "heart of SD-WAN solution." It is the protocol through which all routes are learned by the vSmart controller and subsequently reflected to other edge devices (vEdge routers). OM runs over DTLS/TLS tunnels established between vEdge devices and the vSmart controller.

2. Core Concepts of Routing: BGP as a Parallel

The session emphasizes that understanding OM doesn't require extensive traditional routing knowledge, but parallels are drawn with BGP (Border Gateway Protocol) due to its similar best-route selection mechanisms and attribute-based decision-making.

Key Routing Concepts Explained (using BGP as an example):

- **Best Route Selection:** This process occurs when a router receives the same route from multiple sources or paths and needs to determine the optimal one.
- **Attributes:** Routing protocols like BGP attach "attributes," "qualities," or "tags" to routes to aid in best-path selection. These attributes can be manipulated for traffic engineering.
- **AS Path (Autonomous System Path):** Represents the sequence of Autonomous Systems (AS numbers) a route has traversed.
- The "path length" is the number of AS numbers in the AS path.
- **Rule:** "The shorter the length, the better the route."
- **Traffic Engineering with AS Path Prepending:** By adding "dummy" AS numbers to a route's AS path, its length can be artificially increased, making it less preferred and allowing traffic to be steered over a different, potentially longer, physical path. This demonstrates how default behavior can be tweaked to meet specific design requirements.
- **Local Preference:** An attribute used to prefer a particular exit point from an AS for internal traffic.
- **Rule:** "If it is higher, it is more preferred."
- Allows an administrator to override AS path length preference and direct traffic through a desired exit interface.
- **Administrative Distance (AD):** Used when the same route is learned via **different routing protocols**.
- **Rule:** "The lower the AD, the highest preferable route." (e.g., EIGRP internal AD is 90, OSPF AD is 110, so EIGRP routes are preferred over OSPF).
- **Metric:** Used when the same route is learned via **the same routing protocol** from different sources.
- **Rule:** "The lower the metric, the better the route."
- **Next Hop Reachability:** A fundamental qualification criterion for any route. If the next hop is not reachable, the route is considered invalid and will not be installed in the routing table, regardless of other attributes.

3. OM Routes and Associated Attributes

OM exchanges different types of routes, each with its own set of attributes:

- **Virtual Routes (V-Routes):**
 - Represent "all the routes at your service side" or "service VPNs" (e.g., corporate, guest VPN routes, connected routes).
 - **Key Attributes:** **T-Log (Next Hop):** Analogous to BGP's next hop. If the T-Log is unreachable, the V-Route is not qualified.
 - **VPN ID:** Identifies which VPN the route belongs to (e.g., VPN 10, VPN 20).
 - **Preference:** Exactly like BGP's local preference. Higher preference means more preferred. Used to manipulate traffic paths.
 - **System IP:** The System IP of the vEdge that originated the virtual route.
 - **Origin Protocol:** Indicates the original routing protocol from which the route was learned (e.g., OSPF, RIP, Static). Important for tie-breaking decisions (e.g., OSPF is generally better than RIP).
 - **Origin Metric:** The original metric of the route from its originating protocol.
 - **Site IDs:** Identifies the site from which the route originated. (Further details to be discussed later).
- **T-Log Routes:**
 - Analogous to next-hops in traditional routing; they define **where to build IPsec tunnels**.
 - OM treats virtual routes and T-Logs separately to provide "more fine control over our topology" because "IPSEC tunnels are going to be formed on the basis of TLOCs. If you control those TLOCs, you can control your IPsec."
 - **Key Attributes:** **Color:** Identifies the transport link (e.g., Internet, MPLS).
 - **System IP:** The System IP of the vEdge associated with the T-Log.
 - **Encapsulation Type:** (e.g., IPsec, GRE).
 - **Preference:** Preference specifically for the T-Log itself, distinct from the V-route preference.
 - **Weight:** Important for load balancing. Allows for unequal cost load balancing (e.g., 30% traffic over one T-Log, 70% over another).
 - **T-Log Private Address / T-Log Public Address:** Private and public IP addresses associated with the T-Log.

4. OM Path Selection Algorithm

The OM path selection algorithm is "similar with whatever we have in BGP." It follows a specific ordered evaluation of attributes to select the best path:

- **Select Active and Valid OM Routes:** **Valid:** Next hop (T-Log) must be reachable.
 - **Active:** Routes directly learned from the vSmart controller are preferred over "graceful restart" routes (routes retained from a previously active vSmart that might now be down).
1. **Prefer Locally Sourced OM Route:** If a vEdge router has a locally originated route, it will prefer that over a route learned from the vSmart controller.
 2. **Prefer Route with Lower Administrative Distance (AD):** If AD values are configured, the route with the lower AD is preferred.
 3. **Prefer Route with Highest OM Preference:** This refers to the **V-Route preference** attribute. The route with the highest preference value wins.
 4. **Prefer Route with Highest T-Log Preference:** If the OM preference is the same, the algorithm then considers the **T-Log preference** attribute. The T-Log with the highest preference wins.
 5. **Highest Origin:** If previous criteria are same, the origin protocol (e.g., Connected > OSPF > RIP) is used as a tie-breaker.
 6. **Origin Metric:** If origin is also the same, the origin metric is considered.
 7. **(Continues):** The algorithm continues evaluating other parameters until a single best route is determined or if multiple routes are deemed "equal," leading to equal-cost load balancing (if configured).

Key Insight: Policies configured via vManage allow administrators to set these OM/T-Log preference values, influencing the best-path selection algorithm.

5. Traffic Engineering in SD-WAN (Practical Examples)

Traffic engineering in SD-WAN involves manipulating OM attributes to steer traffic according to business requirements, similar to BGP. The **vManage** acts as the single point for defining network intent and pushing policies to the **vSmart controller**, which then instructs the vEdge devices.

Use Case 1: Preferring MPLS over Internet Transport for a Branch

- **Requirement:** For the Delhi branch, always prefer MPLS as the primary transport and Internet as the secondary.
1. **Mechanism:** Log into vManage.
 2. Configure a policy that instructs the vSmart controller.
 3. The policy increases the "Preference" (V-Route preference) for virtual routes learned over the MPLS T-Log when advertising them to the Delhi branch.
 4. Example: MPLS-learned routes get a preference of 500, while Internet-learned routes get a default preference of 0 or 100.
 5. The vEdge in Delhi receives these routes and, applying the OM best-path algorithm, installs the higher-preference MPLS route, directing traffic over the MPLS tunnel.
- **Result:** Traffic from Delhi to the data center will primarily use the MPLS tunnel. If MPLS goes down, the Internet tunnel (secondary) will be used.

Use Case 2: Primary/Secondary Data Centers (Default Route Preference)

- **Requirement:** Designate Data Center 1 (DC1) as the primary for Internet access and Data Center 2 (DC2) as the secondary/DR.
1. **Mechanism:** Both DC1 and DC2 advertise a default route (0.0.0.0/0).
 2. Branches (e.g., Delhi) learn these default routes from both data centers via their respective T-Logs (e.g., T1, T2 from DC1; T3, T4 from DC2).
 3. Log into vManage.
 4. Configure a policy instructing the vSmart controller to increase the "Preference" (V-Route preference) for the default route learned from DC1 (e.g., to 500) when advertising it to branches. DC2's default route retains a lower preference (e.g., 0).
 5. Branches install the DC1 default route due to its higher preference.
- **Result:** All Internet-bound traffic from branches will primarily egress via DC1. This policy can be scaled to apply to hundreds of branches simultaneously using Site IDs.

Important Note on Tunnel Behavior: By default, even secondary or backup tunnels (IPSec tunnels between vEdges) remain "up" and established, consuming resources. This is analogous to a physical link staying up even if traffic is primarily routed over another. However, "on-demand tunnels" are a feature that can be configured to bring up tunnels only when traffic is present, optimizing CPU utilization for vEdge routers, especially in large deployments.

6. SD-WAN and Legacy Routing Parallels

The session repeatedly emphasizes that SD-WAN does not reinvent routing fundamentals but rather provides a centralized and simplified way to manage them.

- **Redistribution:** Just like in traditional networks, redistribution is required at the vEdge between the OM domain and the local LAN routing protocols (e.g., OSPF, BGP).
- **Routing Loops:** SD-WAN does not eliminate routing loops. Network engineers still need to implement mechanisms like route tagging and denial (supported in SD-WAN) to prevent loops during mutual redistribution between OM and other protocols.
- **Intent-Based Networking:** SD-WAN's primary advantage is its "intent-based" approach, where policies are defined centrally in vManage and pushed to the vSmart, which then orchestrates the network, greatly simplifying configuration and scaling compared to traditional per-device configurations.

7. OM is Proprietary (Cisco Viptela)

OM is a proprietary protocol developed by Viptela (now a Cisco property after acquisition).

Day 17 & 18:

SD-WAN Fundamentals: OMP, Templates, and NAT Deep Dive

1. Introduction and Session Overview

The session, designated as the 17th and 18th in the SD-WAN journey (despite being the 9th actual class), primarily focuses on a deeper dive into **Overlay Management Protocol (OMP)**, **templates**, and **Network Address Translation (NAT)**, with a potential discussion on Direct Internet Access (DIA) if time permits. The core agenda includes:

- Demonstration of OMP and templates within vManage.
- A comprehensive understanding of NAT and its implications for network tunnels.
- Discussion of DIA in the context of SD-WAN, emphasizing the prerequisite understanding of NAT.

The instructor reiterates the importance of understanding NAT, stating, "believe me guys network address translation understanding network address translation uh that why if i implement somewhere NAT in my uh complete network then there are some channels which are not coming up right and believe me i see these questions from lot of guys in real production right that i have done everything but some of my tunnels are not coming up and believe me the problem is not with respect to solution itself. The problem is with respect to net understanding."

2. Lab Setup and Edge Device Bring-Up Procedure

A significant portion of the session is dedicated to the practical aspects of setting up the SD-WAN lab, specifically bringing up the vManage, vSmart, vBond, and Linux machine.

2.1 Core Components and Authentication

- **Essential Components:** The vManage, vSmart, vBond, and a Linux machine (for vManage UI access) must be "up and running."
- **Authentication and Certificates:** These devices must authenticate each other, which relies on **certificates**. vManage is designated as the Certificate Authority (CA) and signs all certificates.
- **NTP Synchronization:** **Network Time Protocol (NTP)** synchronization is "the crucial factor" for certificate authentication. Mismatched times can lead to certificate authentication failures, preventing vManage, vSmart, and vBond from functioning. As the instructor noted from personal experience, "i have seen cases where your certificate authentication is failing and if your certificate authentication fails, basically these device are not going to authenticate each other, which is going to result into your lab. uh where v manage vs smart and vbond are not working right."

2.2 Bringing Up Edges (VH1, VH2, VH3)

The recommended approach is to bring up edges connected to the internet first, gradually adding complexity:

- **Internet Cloud and Static Routes:** The internet cloud and a simple router with static routes need to be configured first.
- 1. **Edge Configuration Steps:**
 - Initial Configuration:** This includes system IP, interface IP addresses, color (e.g., "biz internet"), encapsulation (IPsec by default), and NTP server settings. These are provided in shared configuration files.
 - Certificate Installation:** Since these are virtual devices, the root CA certificate (from vManage) needs to be manually installed on each edge. This is crucial for the TLS/DTLS channel between the edge and vBond.
 - Registration Request:** The edge initiates communication with vBond, sending its chassis ID or serial number (e.g., 123456). vBond authenticates this against its pre-uploaded list of authorized serial numbers. If the numbers match, vBond provides the IP addresses of vManage and vSmart to the edge.

2.3 Control and Data Plane Connections

- **vBond's Role:** vBond acts as the "first step of bringing up all these edges together," it authenticates the edge and provides the vManage and vSmart IP addresses. The DTLS tunnel with vBond is temporary and "is going to be removed once you are done with the vBond."
- **Persistent Tunnels:** Once the edge receives the vManage and vSmart IP addresses, it establishes persistent control connections with them. The vManage dashboard typically displays only the OMP connections (between edge and vSmart).
- **T-Lock Information:** After OMP tunnels are up, edges begin sending **T-Lock information** (system IP, color, encapsulation) to vSmart. vSmart learns these T-Locks and advertises them to other edges. This enables edges to establish IPsec data tunnels between themselves.
- **IPsec Tunnels and BFD:** IPsec connections are formed between T-Locks. **Bidirectional Forwarding Detection (BFD)** runs over these IPsec tunnels to calculate performance metrics (delay, jitter, packet loss), which are used in routing applications.

2.4 Verifying Connectivity and OMP Routes

- **vManage Dashboard:** The vManage main dashboard provides a "single pane of glass" view, showing controller status (vSmart, vEdges, vBond, vManage), certificate warnings, and control/data plane connectivity.
- **Show Commands on vSmart:** show omp peers: Shows OMP peers (connected vEdges).
- **show omp routes:** Displays OMP routes, including T-Locks and virtual routes (connected/OSPF intra-area routes from the LAN side). These routes show attributes like site ID, originator, T-Lock (system IP, color, encapsulation), preference, and origin protocol.
- **Show Commands on vEdge:** show control connections: Lists connections to vManage, vBond, and vSmart.
- **show ip routes:** Shows connected routes (VPN 10 in the example) and routes learned from other sites via OMP.
- **show bfd sessions:** Displays IPsec tunnels, as BFD runs over them to monitor performance.

2.5 Troubleshooting and Learning

The instructor emphasizes that encountering problems during lab setup is an opportunity for learning. "If you are doing wrong things. If you're doing if your lab is not coming up, if you are facing problems, there is the opportunity to learn." Debugging logs and understanding the underlying processes (like certificate authentication issues due to timing) are crucial for deep learning.

3. Network Address Translation (NAT) Deep Dive

NAT is explained as a critical concept, particularly for troubleshooting SD-WAN tunnels.

3.1 Private vs. Public IP Addresses

- **Private IP Ranges:** 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16. These are "not routable over internet or any service provider medium."
- **Need for NAT:** To communicate with the internet, private IP addresses must be translated to public IP addresses.

3.2 Static NAT (One-to-One NAT)

- **Concept:** A fixed, one-to-one mapping between a private IP address and a public IP address.
- **Mechanism:** When a packet from an inside host (e.g., 10.2.10.10) goes out, the NAT router changes its source IP to a public IP (e.g., 20.20.20.10). When a reply comes back to the public IP, the NAT router translates the destination IP back to the private IP.
- **Bidirectional Communication:** "It doesn't matter who is initiating the traffic." Both inside-to-outside and outside-to-inside communication is possible because the static mapping is "always there."
- **Limitation:** Not scalable for large numbers of users as it requires a unique public IP for each private host.

3.3 Port Address Translation (PAT) / NAT Overload

- **Concept:** Allows multiple private IP addresses to share a single public IP address by differentiating flows based on source and destination port numbers.
- **Mechanism:** When an inside host initiates communication, the NAT router translates the source IP to its public IP and also changes the source port (ephemeral port). This creates a dynamic entry in the NAT table that includes the source IP and port.
- **Unidirectional Initiation:** "The communication between this user and server can happen only and only if this user is initiating the communication." If an outside host attempts to initiate communication, there is no corresponding dynamic entry, and the packet will be dropped. "If somebody coming from outside they do not know how to communicate. This is the catch here and this is the most important part you need to understand."

3.4 NAT and SD-WAN Tunnels

- **Problem with Symmetric PAT:** If **Port Address Translation (PAT) / Symmetric NAT** is enabled on **both sides** of an IPsec tunnel (e.g., between two vEdges behind separate firewalls doing PAT), the tunnel "is not going to come up." This is because IPsec tunnel formation often involves one side initiating the connection, and the other side (behind PAT) may not have an active session to allow the return traffic. The instructor states, "If symmetric nat is there at the both the side it is not going to work.. If both symmetric, there is a problem. There is not going to be any IP sector."
 - **Solution:** At least one side of the connection should have **Static NAT** configured or no NAT at all.
1. **How vEdges Learn Public/Private IPs (VBond's Role):** When a vEdge behind a NAT device connects to vBond, it sends a DFLS message.
 2. The vEdge's configured (private) IP address is included in the message data.
 3. The NAT device translates the source IP of the packet to its public IP.
 4. vBond receives the packet, opens it, and observes both the public source IP (from the packet header) and the private IP (from the message data).
 5. vBond identifies that the vEdge is behind a NAT device and informs the vEdge about its public IP.
 6. The vEdge then includes both its private and public IP addresses in the T-Lock information advertised to vSmart.
 7. vSmart disseminates these T-Lock details to other vEdges, allowing them to initiate IPsec tunnels to the correct public IP addresses of their peers.
- vBond acts as a **STUN (Session Traversal Utilities for NAT) server**, helping edges discover their public IP addresses.

3.5 SD-WAN Topology and NAT Best Practices

- **Full Mesh vs. Hub-and-Spoke:** If a **full-mesh topology** is deployed and **PAT is enabled on all vEdges**, IPsec tunnels between vEdges will likely not come up due to the symmetric NAT problem.
- **Hub-and-Spoke Solution:** In a **hub-and-spoke topology**, if the hub (e.g., data center vEdge) uses Static NAT or no NAT, and spokes use PAT, tunnels can be formed. The instructor suggests having "a static nat here and it doesn't matter what you have here maybe back maybe somewhere anywhere anything."
- **Remote Branch Deployments:** In remote branches with few users, the SD-WAN edge device might directly face the internet, eliminating the need for NAT. However, in data centers, vEdges are typically behind firewalls, which may perform NAT.

4. Practical Demonstration with vManage

The instructor demonstrates real-time verification of OMP peers, OMP routes, and T-Locks on the vSmart and vEdge devices via the vManage UI.

- **vManage Dashboard:** Confirms three vEdges, one vBond, one vManage, and one vSmart are up, with control connections established.
- **vSmart CLI (via vManage):** `show omp peers`: Confirms peering with vEdges (note 100, 200, 1000 DC).
- `show omp tlo paths`: Shows learned T-Lock paths for all three edges, including system IP, color, and encapsulation.
- `show omp routes`: Displays service-side routes (e.g., 10.1.1.0/24 for VPN 10) learned from vEdges, along with their associated OMP attributes (originator, T-Lock preference, tag, etc.).
- **vEdge CLI (via vManage):** `show control connections`: Shows active control connections to vManage and vSmart (vBond connection is torn down).
- `show ip routes`: Verifies connected routes and routes learned from other sites via OMP.
- `show bfd sessions`: Displays active IPsec tunnels to other vEdges, indicating their status (up).
- **vManage UI (Monitor > Network > WAN Tunnels):** Provides a visual representation of WAN tunnels for specific vEdges, showing T-Lock details (color, jitter, loss, latency).

Day 19 & 20:

SD-WAN Core Concepts and Design Considerations

1. Visualization of SD-WAN Tunnels: Analogies to VLANs and Trunks

The session begins by simplifying the abstract concept of SD-WAN tunnels by drawing parallels to familiar networking concepts: VLANs and Trunks.

- **VLANs as Service-Side VPNs:** Just as VLANs segregate traffic within a switch, SD-WAN uses Service-Side VPNs (e.g., VPN 10, VPN 20, VPN 30) to logically separate different user groups or traffic types within an organization (e.g., corporate vs. guest traffic).
- **IPSec Tunnels as Trunks:** The speaker explains that an IPSec tunnel in SD-WAN functions much like a trunk link between switches. A single IPSec tunnel can carry data for multiple Service-Side VPNs, similar to how a trunk carries traffic for multiple VLANs using 802.1Q tagging.
- "If you consider this switch as a router here, right? And if you consider this VLANs as your service-side VPN, maybe VPN number 10, VPN number 20 and VPN number 30, you are not connecting your edges with different links, right? If you need to pass the data between your edges, what you are going to do, you are creating IPSec channels here and that IPSec channel is actually carrying data of different VPNs just like this trunk is carrying data for different VLANs. It is as simple as that."
- **VPN Value in IPSec Header:** The mechanism for carrying multiple VPNs over a single IPSec tunnel involves embedding the "VPN value" within the IPSec header. This allows the destination edge device to correctly forward the packet to the appropriate VPN.
- **Underlay (VPN0) and Overlay:** The speaker clarifies that the IPSec tunnel itself is not part of a specific Service-Side VPN (like VPN 10) nor exclusively part of the Underlay (VPN0). Instead, it exists "somewhere in the middle," facilitating communication between Service-Side VPNs across the Underlay.
- **Route Leaking (Automated in SD-WAN):** SD-WAN automates the "route leaking" process, where routes from Service-Side VPNs (e.g., VPN 10) automatically understand their next hop is in the Underlay (VPN0) to reach other edges. This simplifies configuration compared to traditional routing where manual route leaking between VRFs might be required.
- "The system the solution is doing on behalf of you, because it knows whenever uh the packet comes from VPN 10, VPN 20, it needs to uh send it to the IPC tunnel which is uh which eventually going to use the underlay."

2. Direct Internet Access (DIA)

Direct Internet Access (DIA) is presented as a crucial customer requirement in SD-WAN, offering cost savings and improved user experience by allowing specific traffic to directly exit the local branch without backhauling to a central data center.

- **Traditional Backhauling:** Before DIA, all internet bound traffic from a branch, even if the branch had direct internet connectivity, would typically be routed through an IPSec tunnel to a central data center, often for security inspection by firewalls.
- "Why it has to go at the first place to internet over this tunnel sorry to the data center over this tunnel then it is going to the firewall and then basically going to to reach this 8.8.8.8 right this was the problem we had because by default everything is going over these IP's tunnel and you have form IPSec tunnel with your data center over two transport one is MLS one is Internet although you have a direct connectivity to internet over this branch Delhi but you are not going directly you are going first to data center."
- **Customer Requirement for DIA:** Customers often request DIA, especially for guest users or specific applications (e.g., Office 365), to avoid unnecessary backhauling and improve performance.
- "Your boss came and say I do not want to do that. I want at least for my guest users that they can go directly to internet because I do not want to want the guest traffic to come over MS first and they are going to be scrapped again, my firewall and then they go out. I do not want that."
- **Implementing DIA: Static Routes and NAT:** To implement DIA, a static default route (0.0.0.0/0) is configured within the specific Service VPN (e.g., VPN 20 for guest users) that requires direct internet access. This static route has a lower administrative distance (AD=0) than the default routes learned via OMP (AD=250), making it preferred.
- "You are defining it statically here 0.0.0.0 and now the next hop you are saying it is VPN0 previously The next hop was T-LO T1 and T2 right now you are saying VPN zero and maybe the VPN0 the interface is, suppose this one 0/1 this is your internet interface right and if you say this then because this is a static route and by the way guys they say this route as net da dia route direct internet access and they say that its administrative distance value is six which is again smaller than OMP admin value which is 250."
- **NAT on Underlay Interface:** Since client IP addresses in Service VPNs are often from private ranges, Network Address Translation (NAT) must be enabled on the Underlay (VPN0) interface connected to the internet. This converts private source IPs to publicly routable IPs.
- "To make sure your DIA is working with this route, you need to enable first thing first NAT under the interface of VPN0 which is connected to internet. It is as simple as that."
- **Security Considerations with DIA:** Opening direct internet access at the branch level raises security concerns.
- **VH as a Firewall:** SD-WAN routers (VHs/CHs) possess built-in firewall capabilities (intrusion prevention, URL filtering, application identification) that can be leveraged for security at the branch.
- **Cloud Security (Cisco Umbrella):** For larger branches or high traffic volumes where overloading the VH with firewall tasks is a concern, integration with cloud security solutions (like Cisco Umbrella) is an alternative. This allows traffic to be scrubbed by the cloud security provider before reaching the internet, potentially reducing latency by avoiding backhauling to the central DC firewall.
- "Instead of sending it to anywhere else what you are doing is uh whatever the traffic coming to your office 360 first you are going to send and there is a channel between these two guys right you are sending uh you see this is again the DIA but you are passing you are not sending everything in the clear you are first sending it to your cloud security uh provider maybe it's umbrella and umbrella is something where you can configure your security rules..."

3. TLOC Extension: Cost-Saving Redundancy

TLOC Extension is introduced as a cost-saving mechanism for achieving redundancy by allowing one edge device to extend its transport connectivity to another edge device within the same site.

- **Problem:** Traditional full redundancy across multiple WAN connections (e.g., MPLS and Internet) at a single site would typically require four separate circuits (two MPLS, two Internet). This increases cost.
- "At the end of the day you are buying four circuits from the service provider... Can we reduce these four circuits from four to two?"
- **Solution: TLOC Extension:** TLOC Extension allows two edge devices at the same site to share their underlying transport connections. For example, if VH1 has MPLS and VH2 has Internet, TLOC extension can enable VH1 to utilize VH2's Internet connectivity and vice versa, creating full redundancy with fewer direct service provider connections to each device.
- "I am just taking two more interfaces. So previously if you talk about 0/1 interface it is connected to MLS. It means that your VH1 has MS connectivity no doubt but VH1 or H1 does not have internet connectivity. And how how I'm providing it just try to understand I'm taking another interface in VPN0 right here this one 0/3 interface and connecting it to edge2 right here again this interface is also part of VPN0 right and system is doing some magic and it is connecting these two interfaces in the device itself."
- This is achieved by connecting a LAN interface from one edge device (in VPN0) to another edge device (also in VPN0) to extend the reachability of a transport type.
- **Common Production Issue: Route Advertisement:** A common pitfall in TLOC extension is forgetting to advertise the routes of the extended transport's subnet into the respective WAN cloud (e.g., advertising the extended MPLS subnet into the MPLS cloud). Without this, the TLOC cannot be reached, and tunnels will not form.
- "You forget to advertise this route x.x.0 into mls right in all the settings 1 and everything what you have done everything but you haven't advertised this route into mpls you have to have you have to advertise this route out otherwise how this guy should come to know about where my x.x.x.0 is."
- **Same Site Restriction:** TLOC Extension can only be configured between edge devices within the same site.

4. Tunnel Control: Restrict Keyword and Tunnel Groups

SD-WAN provides mechanisms to control which tunnels form between TLOCs, crucial for optimizing resource utilization and aligning with network design.

- **Default Behavior (Full Mesh):** By default, SD-WAN aims for a full mesh of tunnels, meaning every TLOC attempts to form tunnels with every other TLOC it learns from the vSmart.
- "As you know the default behavior is going to be full mesh. So what is going to happen? Everybody is sharing its route with this VS smart. VS smart is letting know this all the edges regarding other transports."
- **Problem with Default Behavior:** If transports are not directly reachable (e.g., MPLS TLOCs trying to form tunnels with Internet TLOCs when there's no cross-connectivity), CPU cycles are wasted attempting to form non-viable tunnels.
- "This device at least some CPU cycles are wasted informing that those channels which are not going to be come up."
- **restrict Keyword:** The restrict keyword, configured on an interface along with its color (transport type), instructs the device to only form tunnels with other TLOCs that have the same color. This is useful when different transport types (e.g., MPLS and Internet) are completely separate.
- "You just need to say restrict. What is the meaning of it? You are saying restrict your tunnel making to only blue colors."
- **Tunnel Groups:** Tunnel Groups offer more granular control than restrict. They are used when different transport types should form tunnels, even if they have different colors, because they are logically part of the same "group" of tunnels (e.g., Metro Ethernet, a Layer 2 VPN service from the same MPLS provider, or different Internet service providers).
- "Because metro ethernet is part of my MLS and I do want that this guy H1 should create a tunnel with me also which it is not creating it is the problem for me right right? Because color is different."
- By assigning interfaces to a "tunnel group" (e.g., tunnel group one), tunnels can form between TLOCs of different colors if they belong to the same group. This is essential for scenarios where different transports are conceptually linked.
- "Instead of changing the color which you cannot here you can configure this interface is also part of tunnel group one. And now you are not dependent on the restrict.. You have mentioned that every interface whatever you are seeing here is a part of tunnel group one. And if that is the case, this guy is going to create a connection like this also. All three tunnels are going to be up and running this way."

5. Management and Monitoring

- **vManage Centralized Control:** vManage is the central point for managing SD-WAN devices, pushing configurations to edge devices.
- **PNP and Startup Configuration:** Edge devices typically boot in a "Plug and Play" (PNP) stage, automatically discovering and connecting to the vBond, then pulling configurations from vManage. Minimal startup configurations can be manually applied to guide this process if needed. Once managed by vManage, devices are centrally controlled.
- **Monitoring Tools:** SD-WAN offers robust monitoring capabilities:
- **vAnalytics:** Provides insights into application usage over tunnels.
- **BFD (Bidirectional Forwarding Detection):** Monitors delay, jitter, and packet loss on interfaces.
- **Graphical User Interface (GUI):** Allows monitoring of live tunnel status and traffic usage.
- **Cisco ThousandEyes:** Integrated with SD-WAN to provide advanced network visibility and monitoring, including path visualization.

6. Future Topics and Outlook

- **Application-Aware Routing:** The next session will cover Application-Aware Routing, a crucial feature that allows SD-WAN to dynamically select the best path for applications based on real-time network conditions (delay, jitter, packet loss).
- **Design Phase Emphasis:** The speaker repeatedly stresses that understanding these core concepts, combined with a strong grasp of traditional routing, is essential for the upcoming "design phase" of the course, where customer requirements will be translated into SD-WAN solutions.
- "From now on whatever we are learning those are the direct requirements from the customer which you are going to see if you are building customer network."
- "Routing is exactly same your mind your thought process is going to be absolutely same."
- **Security Integration:** The session highlights the growing importance of security knowledge for network engineers, as security features are increasingly built directly into networking devices.
- **Cloud vs. On-Premise Data Centers:** The speaker acknowledges the trend towards cloud adoption but also notes that the initial cost-saving motivation for cloud migration is not always realized, suggesting data centers will likely remain a part of many networks.

Day 21 & 22:

SD-WAN Fundamentals and Application-Aware Routing: A Detailed Briefing

1. SD-WAN Fundamentals: Setting the Stage

The session emphasizes that while advanced policy design will be covered later, understanding foundational concepts is crucial for effective SD-WAN implementation. These fundamentals underpin more complex features like Application-Aware Routing.

- **Policy Design in Later Stages:** "Believe me your policy is design. So we are going to cover Maximum part of policy when we talk about design when we are converting a customer requirement into a high level design or the low-level design that is where we are going to discuss most of the policy section."
- **Importance of Planning:** The planning phase is paramount for any project, including SD-WAN deployments. "But believe me the most important part in implementation of a project or in starting a project from the scratch is the planning part. And I could see that people do not spend time here. And if you do not spend time in the planning section, believe me, the design and implementation part is going to be difficult." This includes understanding the customer's existing network and future expansion plans.
- **Moving from Fundamentals to Design:** The course is transitioning from basic concepts to the design aspects of SD-WAN, including policy, cloud integration, and migration strategies.

2. Understanding Network Performance Metrics: Loss, Latency, and Jitter

Application-Aware Routing (AAR) critically depends on accurately measuring network performance. The briefing emphasizes three core metrics: packet loss, latency, and jitter.

Key Concepts:

- **Packet Loss:** Occurs when packets are discarded by network devices, often due to an overloaded network or bandwidth issues.
- **Calculation:** Measured as the percentage of discarded packets out of the total sent. For accurate calculation, a "large number of inputs or large number of packets" should be used. For example, "If we have hundred of packets on which we are going to calculate the loss and only three or four packet is dropped by this guy. Right now we can say the packet loss is only 4%."
- **Impact:** Even small percentages of loss can severely impact application performance (e.g., 20% loss on 10 packets is significant).
- **Latency:** The time taken for a packet to travel from a source (client) to a destination (server).
- **Components:** Includes processing time at routers and transmission time over the wire.
- **Roundtrip Time (RTT):** The total time for a packet to travel from source to destination and back. RTT can be higher than single-direction latency if the return path is different or experiences more delays. "So in the forward direction it is 1 second and it is not the case that it is going to be 1 second in the reverse direction also and the reason is that reverse direction can be of a different path."
- **Jitter:** The variation in latency between consecutive packets.
- **Impact:** Highly problematic for real-time applications like voice (VoIP), as inconsistent packet arrival times lead to a "not going to get a smooth experience."

3. Application-Aware Routing (AAR): A Flagship SD-WAN Feature

AAR is highlighted as a "flagship feature of SD-WAN solution," addressing the limitations of traditional routing protocols in meeting application-specific performance requirements.

Core Principles:

- **Application-Specific SLAs:** Many applications have strict requirements for packet loss, latency, and jitter. For example, a voice application might require "packet loss of more than 4%," "latency between the client and the server is greater than 200 milliseconds," or "jitter value should not be more than 20 milliseconds."
- **Traditional Routing Limitations:** Legacy routing protocols (BGP, OSPF, EIGRP) are not designed to consider these performance metrics. "BGP is not designed to see this BGP is not designed to see the packet loss and on the basis of packet loss it is going to change from packet number one to what circuit number two because the BGP P neighbor is up and running no problems with respect to that." They only focus on link up/down status.
- **Business Criticality:** For "business critical application[s]," meeting these SLAs is crucial to avoid "business loss."
- **Dynamic Path Selection:** SD-WAN can "map an application to a particular tenor" and dynamically shift traffic between different transport tunnels (e.g., MPLS, Internet, LTE) to ensure application SLAs are met. If a primary circuit (e.g., MPLS) fails to meet an application's requirements (e.g., due to increased packet loss or latency during peak times), SD-WAN automatically reroutes traffic to an alternative tunnel that is currently performing within the specified SLAs.
- **Decision-Making:** Decisions to shift traffic are not impulsive. They are based on "app pole interval" (default 10 minutes) and an "app multiplier" (default 6), meaning the system monitors average performance over a period (e.g., 1 hour by default) before making a decision. This prevents "impulsive" shifts due to temporary spikes.
- **Configurable Actions:** If no tunnel meets the SLA, administrators can configure various actions, such as reverting to normal routing, dropping packets (strict action), or staying on the primary link despite degraded performance.

4. Bidirectional Forwarding Detection (BFD)

BFD is presented as a critical enabler for Application-Aware Routing, providing rapid link failure detection and performance monitoring.

Key Functions:

- **Rapid Failure Detection:** BFD offers "subsecond failover" by rapidly detecting link failures, much faster than traditional routing protocols' keep-alive mechanisms (e.g., BGP's 180-second dead timer).
- **How it Works:** BFD sends "BFD eco packets" at frequent "hello time" intervals (e.g., every 1 second). If a configured number of consecutive packets (e.g., three, determined by the "multiplier" parameter) are missed, BFD quickly informs the routing protocol (e.g., BGP) to bring down the neighbor relationship. "So R1 through BFD can know in 3 seconds instead of 180 seconds right I'm talking about BFD now that it knows in 3 second that there is some problem with the link and it can inform the BGP."
- **Performance Measurement:** Beyond liveness detection, BFD packets can be leveraged to "calculate all these things like packet drop, you can calculate latency, you can calculate jitter." This data is essential for AAR.

5. Collecting Customer Requirements for SD-WAN Design

The briefing outlines a structured approach for network consultants to collect requirements from customers looking to transition to SD-WAN.

Key Steps and Questions:

- **Understand Existing Network:** This is the "first question." Consultants must thoroughly understand the "legacy network," including WAN and LAN infrastructure, routing protocols, and traffic flows. "If you do not understand the existing network well there is going to be problem with respect to implementation of SDA or moving this legacy network from legacy to SDA."
- **Identify Current Issues:** Determine "why customer is moving from legacy to SD wan." This includes problems like network agility ("network is actually not agile enough"), single points of failure, application performance issues not detectable by legacy protocols, and "nightmare" of current network management (lack of "innovation").
- **Future Expansion Plans:** Inquire about "future expansion plan[s]" including new branches, data centers, or geographical expansion, to ensure the SD-WAN design is scalable.
- **Cloud Presence:** Understand current or planned "cloud presence," including whether data centers are moving to the cloud, and if single or multiple cloud providers are targeted.
- **Security Policies:** Review existing security policies, typically enforced at data centers, and discuss the customer's desire to extend "the same kind of security to be implemented at the branches" directly to the internet (local internet breakout). SD-WAN edges can act as "full-fledged firewall[s]."
- **Network Segmentation:** Ask about "guest network[s]" or "VRFs" for different purposes (e.g., corporate, guest, IoT devices).
- **Migration Strategy and Downtime:** Since most deployments are "brownfield" (converting an existing network), discuss how to migrate from legacy to SD-WAN and the "how much downtime this company can take."

- **Controller Deployment Model:** Determine the customer's preference for controller deployment: "on-prem" (customer manages) or "Cisco services in Cisco's cloud" (Cisco manages).
- **On-prem:** Preferred by organizations with strict data sovereignty or security compliance (e.g., banks, government), but requires customer or partner to manage.
- **Cloud:** Cisco-recommended, managed by Cisco, potentially offering operational cost savings and faster service delivery. Can be dedicated resources in public clouds (AWS, Azure, GCP).
- **Topology Preference:** Discuss desired network topology (full mesh vs. hub-and-spoke) and how it might vary across geographies.

6. SD-WAN Advantages Over Legacy Solutions (e.g., DMVPN)

The discussion highlights key benefits of SD-WAN that can convince customers to adopt the technology, even if their existing infrastructure is functioning.

Key Advantages:

- **Single Pane of Glass Management:** SD-WAN offers centralized management through "V-Manage," simplifying network operations compared to managing multiple devices and systems in legacy environments. "If you want a manageable solution, SD wan is there for you. Single pane of glass."
- **Automation and Zero-Touch Provisioning (ZTP):** SD-WAN simplifies new site deployments through automation and "plug and play" capabilities, where devices automatically take configurations. "Even you do not need to do anything on this device. Just power it on and it is going to take because of the infrastructure."
- **Advanced Analytics and Assurance:** SD-WAN platforms (e.g., V-Analytics/Assurance) provide "meaningful information out of this data," aiding in performance reporting and troubleshooting.
- **Simplified VPN Management:** Eliminates the need for complex key exchange protocols like IKE and key servers, as IPsec tunnels are built using secure communication over control plane (DTLS/TLS) with V-Manage/V-Smart.
- **Application-Aware Routing:** The core differentiator, enabling dynamic traffic steering based on real-time application performance, a capability absent in legacy routing.
- **Complementary, Not Replacement:** SD-WAN often "complement[s] MS" rather than fully replacing it, offering cost savings by allowing for cheaper internet circuits where appropriate.

Day 23 & 24:

Cisco SD-WAN Implementation Briefing Document: Sports India

1. Executive Summary

This briefing document outlines the key requirements, current network challenges, and a high-level design proposal for Sports India's transition to a Cisco SD-WAN infrastructure. Sports India, the largest sporting goods retailer in India with over 600 branches in 100 cities and a main campus in Mumbai with two data centers (Mumbai and Chennai), is seeking a modern WAN solution to address current performance issues, support global expansion, and enhance operational efficiency. The proposed SD-WAN solution aims to provide an agile, scalable, and secure network that integrates seamlessly with their existing Cisco SD-Access LAN infrastructure.

2. Customer Overview: Sports India

- Industry:** Sporting goods retailer
- Geographic Presence:** Largest in India, with over 600 branches in 100 cities. Expanding to Sri Lanka and planning expansion into the EMEA (Europe, Middle East, and Africa) region, as well as Singapore and Malaysia.
- Headquarters:** Mumbai
- Data Centers:** Two data centers, one in Mumbai and one in Chennai.
- Current WAN Infrastructure:** Primarily an MPLS-dependent, "pretty flat network." Some larger stores have both MPLS and internet connections, but internet links are not actively utilized and only come up when MPLS is unavailable.
- Existing LAN Technology:** Cisco Software-Defined Access (SD-Access).
- Cloud Presence:** Significant adoption of cloud services, with critical applications and workloads moved to Azure and AWS (multi-cloud environment).
- Key Personnel Involved:** Mr. Ashish Chedda (Chief Information Officer)
Mr. Ashish Dagat (Chief Architect of Network Technology)

3. Current WAN Challenges (Pain Points)

Sports India faces several critical issues with its existing WAN infrastructure that impede business operations and employee productivity:

- Incapability to Meet Business Requirements:** The current WAN structure is "incapable as of now delivering the requirements which are needed to run the business."
- Poor Application Experience, Especially for Cloud Applications:** Employees accessing critical business applications (especially those moved to Azure and AWS) from remote sites experience "delays or some jitter or something like that kind of...poor experience."
- Frequent Outages and Business Losses:** "Several outages as well which have resulted in huge business losses." This is often due to link impairments rather than complete link failures, where the network "is not agile to handle such use cases."
- Inefficient Bandwidth Utilization:** Despite "invested huge amounts in purchasing different circuits and highspeed circuits," there is "no return in investment." Internet transport is often "not utilizing the bandwidth properly."
- Lack of Centralized Visibility and Management:** "We don't have single pane of glass from where I could manage my complete main infrastructure." This lack of visibility makes it "very difficult to track my application traffic over van transport," hindering optimization and troubleshooting.
- High Troubleshooting Time for Networking Staff:** Networking staff spend "40% of their time in troubleshooting the network or network issues," leaving less time for "innovating the network."
- Security vs. Performance Trade-off for Cloud Access:** For security and compliance, "all site can access internet or cloud only from nearest data centers." This "traffic path creates a bad experience for the users trying to access any cloud resource."
- Strict Application SLAs Not Met:** Some applications "have strict requirements in terms of latency [and] jitter." The current MPLS provider "is not giving those SLAs," leading to outages even when links are technically "up but it is not working with the desired SLAs."

4. Customer Requirements (Desired Outcomes)

Based on discussions with Sports India's CIO and Chief Architect, the primary requirements for the new SD-WAN solution include:

- Agility and Scalability:** Looking for an "agile and scalable van solution that can meet the current...business requirements as well as the future requirements like if we expand that network should able to expand across the world."
- Enhanced Cloud Connectivity ("Cloud-First Approach"):** Seamless and optimized connectivity to multi-cloud environments (Azure, AWS) with improved user experience. This implies a need for "direct internet access" from branches while maintaining security.
- Improved Application Performance:** Prioritization and optimized routing for critical applications, ensuring strict SLAs are met. This points towards "application aware routing."
- Operational Efficiency & Innovation:** A solution that enables networking staff to "spend more time in innovating the network rather than troubleshooting the petty issues." This includes a "single pane of glass" for management, enhanced "troubleshooting and analytics," and improved "visibility into the network."
- Effective Utilization of Transport Links:** Maximizing the use of all available WAN links (MPLS and internet).
- Data Center Redundancy & Migration:** Support for their two data centers and the plan to "move our on-prem data centers to collocation facilities to reduce operating expense."
- SD-Access and SD-WAN Interoperability:** The WAN solution "could interoperate with my LAN infrastructure" (Cisco SD-Access).
- Security Compliance with Direct Internet Access (DIA):** Ability to "go directly to cloud from remote center without compromising on my security policies." This might involve "service firewalls at local branches or crowd security."
- Regional Breakout:** Potential for regional breakouts for data traffic to support global expansion.
- High Availability:** Redundant WAN edges and consideration for site categorization to ensure high availability.
- Quality of Service (QoS) / Application Prioritization:** "Prioritize some applications" to ensure optimal performance for critical services.
- Packet Loss Protection:** Desire to protect business-critical applications from packet loss (e.g., forward error correction).
- Potential Multicast Support:** Although not explicitly stated by the customer, a potential future requirement or advanced need.

5. High-Level Design (HLD) Considerations

This section outlines the initial high-level design considerations to meet Sports India's requirements. Further detailed discussions with their network engineering team will refine these aspects.

5.1. Controller Deployment Model

- Recommendation:** Cisco Cloud-Hosted Controllers.
- Rationale:** Aligns with Sports India's "cloud-first approach" and their goal to "reduce operating expense" by moving to collocation facilities. It also minimizes the operational burden on their networking staff, allowing them to "innovate the network rather than troubleshooting the petty issues." Cisco takes responsibility for hardware, certificates, and communication.
- Alternative (On-Premise):** While technically feasible (and what we've practiced in the lab), this would require Sports India to manage the virtual machines and certificates, adding to their existing staff's burden, which they aim to reduce.
- Redundancy:** Minimum of two vSmart controllers for redundancy (horizontal redundancy). These can be deployed in different availability zones or regions to enhance resilience.
- At least two vBond orchestrators for initial fabric bring-up redundancy.
- At least one vManage instance (or redundant vManage if the customer requires active-standby database synchronization for very high availability of management plane).
- Global Scale:** As Sports India expands to EMEA, additional vSmart controllers can be deployed in new regions to facilitate horizontal scaling, with full-mesh TLS connectivity between vSmarts in different regions to ensure consistent routing information.

5.2. WAN Edge Devices and Interfaces

- **Current State Assessment:** Critical to perform an inventory audit of Sports India's current WAN edge routers (e.g., ASR 1000-HX series) to determine if they can be upgraded to support Cisco SD-WAN code (e.g., XE SD-WAN image) or if new hardware is required.
- **Device Selection:** If new devices are needed, selection will be based on:
 - **Bandwidth Requirements:** To support current and future expansion.
 - **Throughput (especially IPsec):** SD-WAN encrypts traffic, which impacts performance. Data center routers will require higher throughput devices (e.g., high-end Catalyst 8000 series like C8500-T2X, capable of 50Gbps+ IPsec throughput and supporting up to 10,000 IPsec tunnels). Branch offices will require models suitable for their size (small, medium, large).
 - **Tunnel Capacity:** Number of IPsec tunnels required (e.g., 10,000 IPsec tunnels for high-end devices).
 - **Interface Requirements:** Number and type of WAN interfaces (MPLS, Internet, etc.) for VPN 0 (transport) and LAN interfaces for service-side VPNs.
- **VRF Design:** Implement service-side VRFs (VRFs) to segment traffic (e.g., Corporate VRF, Guest VRF).
- For Guest VRF, consider direct internet access (DIA) from the branch to offload data center resources, as they typically do not require central security inspection for corporate resources.
- Corporate VRF may follow centralized breakout policies or controlled DIA depending on security requirements.

5.3. Network Topology and Traffic Flows

- **Regional Full Mesh (Within Region):** Within each geographic region (e.g., India, EMEA), a full-mesh topology will be leveraged for branch-to-branch communication. Cisco SD-WAN inherently supports this by establishing direct IPsec tunnels between vEdge/cEdge devices (OM tunnels).
- **Benefit:** Low latency, direct communication, and efficient path utilization within the region.
- **Inter-Regional Communication: Option 1 (Recommended - SD-WAN End-to-End Visibility):** Traffic between branches in different regions (e.g., India branch to EMEA branch) will route via the respective regional data centers, utilizing SD-WAN IPsec tunnels established between the regional data centers.
- **Traffic Flow:** Branch A (India) -> India DC -> SD-WAN IPsec Tunnel (India DC to EMEA DC) -> EMEA DC -> Branch B (EMEA).
- **Benefit:** Full end-to-end SD-WAN visibility, control, and policy enforcement across the entire path.
- **Option 2 (Customer DC Preference - Limited SD-WAN Visibility):** If Sports India prefers to use existing high-quality DCI (Dark Fiber) links between data centers, SD-WAN tunnels may be terminated at the regional DCs, and traffic can traverse the DCI link unencapsulated by SD-WAN.
- **Traffic Flow:** Branch A (India) -> India DC (SD-WAN tunnel terminates) -> DCI Link -> EMEA DC (SD-WAN tunnel originates) -> Branch B (EMEA).
- **Drawback:** Loss of end-to-end SD-WAN visibility and control over the DCI segment. This approach should only be considered if the customer has strict requirements for their DCI.
- **Internet and Cloud Access (Direct Internet Access - DIA):** Leverage DIA from branches for cloud applications and general internet access to improve user experience and reduce backhauling to data centers.
- **Security Integration:** Implement security services directly on the WAN edge devices (e.g., built-in firewall, IPS/IDS capabilities of Catalyst 8000v/cEdge devices) or integrate with cloud-based security solutions (e.g., Cisco Umbrella) to ensure "security is topnotch" without compromising performance.
- **Application Aware Routing (AAR):** Implement AAR to dynamically steer application traffic over the most optimal path based on real-time network conditions (latency, jitter, packet loss) and application-specific SLAs. This directly addresses the current issues with "impaired" links and poor application experience.
- Prioritize critical business applications to ensure their performance.

5.4. Site Classification and Numbering (Site ID Design)

- A crucial aspect for effective policy creation and management, especially given Sports India's large number of branches (600+).
- **Categorization Examples: Data Centers (Extra Large):** e.g., Site IDs 2000-2999
- **Large Branches:** (e.g., dual MPLS connectivity with two routers); e.g., Site IDs 3000-3999
- **Medium Branches:** (e.g., Internet + MPLS connectivity with one router); e.g., Site IDs 4000-4999
- **Small Branches:** (e.g., Internet-only connectivity with one router); e.g., Site IDs 5000-5999
- **Regional Differentiation:** Incorporate regional identifiers into the site ID scheme (e.g., 1xxx for India, 2xxx for EMEA) to easily differentiate and apply region-specific policies.
- **Benefits:** This systematic site ID approach simplifies the creation of centralized policies in vManage, allowing policies to be applied efficiently to groups of similar sites or specific regions using ranges and commas, rather than individual site IDs.

5.5. Integration with Cisco SD-Access (LAN)

- The chosen SD-WAN solution (Cisco Viptelia) is "totally compatible" with Cisco SD-Access. This ensures a cohesive "LAN and the WAN environment is going to be...totally comparable with each other."
- High-level design will outline the seamless routing integration between the SD-Access fabric (core switch) and the SD-WAN edge device at each campus/branch.

6. Next Steps & Action Items

- **Detailed Network Assessment:** Schedule an on-site meeting with Sports India's network engineering team to:
 - Gather precise details on current application usage, specific SLAs, and current bandwidth utilization per link and per site.
 - Understand existing routing protocols (e.g., BGP configuration) and traffic flows in greater detail.
 - Inventory current WAN edge hardware to determine upgrade feasibility.
 - Clarify cloud connectivity details (e.g., ExpressRoute, Direct Connect) and current user access paths.
 - Confirm existing DCI implementation and preferred use for inter-data center traffic.
- **Low-Level Design (LLD) Development:** Develop a detailed LLD document based on the refined requirements, including:
 - Specific device models, interfaces, and cabling.
 - Detailed VRF configurations.
 - IP addressing schemes.
 - Routing policies and application-aware routing rules.
 - Security policies and integration points.
 - Bill of Materials (BOM).
- **Migration Strategy Planning:** Develop a comprehensive migration plan to transition from the current WAN infrastructure to Cisco SD-WAN, minimizing disruption to business operations. This will include dedicated sessions on migration considerations.
- **Policy Development:** Work with Sports India to define and implement granular centralized and localized policies on vManage to meet their specific business and application requirements.
- **Proof of Concept (PoC) / Pilot Deployment:** Consider a pilot deployment at a few representative sites to validate the design and gather feedback before a full-scale rollout.

Day 25 & 26:

SD-WAN Policies: A Detailed Briefing

1. Introduction to SD-WAN Policies: The Core of Network Design

The central theme of the session is the critical role of SD-WAN policies in designing and controlling a wide area network (WAN). The speaker highlights that while bringing up controllers and edges (the foundational elements of an SD-WAN solution) constitutes only "50 to 60%" of the solution, "SD-WAN policies" are the "most important thing" for designing a WAN network. They enable an administrator to move beyond default full-mesh topologies and create custom network designs.

The speaker aims to demystify SD-WAN policies, asserting that they are "just piece of cake here" and "not difficult at all" if attendees concentrate and understand the underlying concepts.

2. SD-WAN Policies vs. Normal Routing Policies: A Comparative Approach

A significant portion of the session is dedicated to comparing SD-WAN policies with traditional routing policies, particularly in the context of Border Gateway Protocol (BGP). The speaker argues that the perceived difficulty of SD-WAN policies stems from a lack of understanding of "normal routing policies."

Key Comparison Points:

- Underlying Principle:** "SDN is just nothing but the routing," implying that the core concepts of routing (e.g., next-hop manipulation, route filtering) are transferable.
- Three Core Steps:** Both traditional routing and SD-WAN policies follow a similar three-step process:

1. **Identification:** Identifying the routes or elements you want to affect.

- Normal Routing:** Achieved using Access Lists (ACLs) to "identify the route."

- SD-WAN:** Achieved using **Site Lists** (e.g., site-list hub, site-list spokes) to identify sites (groups of edge devices). The speaker states, "Just like access list, the name is changed because SD1 is developed way later than your normal routing."

1. **Logic Definition (Match & Action):** Creating "if-and-else" statements to define actions based on identified elements.

- Normal Routing:** Achieved using **Route Maps**. A route map "can create a log logic like if or else" and allows for actions like set the next hop.

- SD-WAN:** Defined as "**logic**" or "**topology logic**." This involves matching specific TLOCs (Transport Locations) or virtual routes and setting parameters like the next hop TLOC.

1. **Applications:** Applying the defined logic to a specific context.

- Normal Routing:** Applied on BGP neighbor relationships (e.g., neighbor R2 route-map next-hop out), specifying direction (inbound/outbound).

- SD-WAN:** Applied on the **vSmart controller** in a specific direction (outbound being a key example for controlling traffic to spokes). The speaker emphasizes, "Whatever you have created, you are just applying this policy."

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Example Use Case (Hub-and-Spoke Topology):

The session uses the creation of a hub-and-spoke topology out of a default full-mesh as a practical example.

- Problem Statement:** In a full-mesh SD-WAN (where H1, H2, H3 communicate directly), the goal is to prevent direct communication between spokes (H2 and H3) and force traffic to traverse the hub (H1). This means preventing the IPsec tunnel between H2 and H3 from forming.

1. **SD-WAN Solution (Applying the Three Steps):****Identify Sites:** Create site-list for hub (e.g., site ID 100 for H1) and spokes (e.g., site IDs 1 for H2 and 2 for H3).

2. **Define Logic (on vSmart):** Instruct the vSmart, "Whenever you advertise routes to spokes, just advertise or just permit TLocs of your H1 or your DC (Data Center, which is the hub) and nothing else." Additionally, for virtual routes (LAN-side routes), "permit all the virtual routes... but set their next hop to data center TLD or hub TLO." This ensures that spokes only learn the TLOC of the hub (T1) and any routes destined for other spokes will have their next hop re-written to T1, forcing traffic through the hub.

3. **Apply Policy:** Apply this logic on the **vSmart** towards the spokes in the **outbound** direction. "You are instructing Mr. vSmart that whenever you are advertising these routes to my spokes... just advertise accordance to this policy or this logic."

3. The Role of vSmart and vManage in Policy Implementation

The session clarifies the hierarchy and interaction between vManage, vSmart, and the edge devices in policy application.

- vSmart: The "Smartest Guy" in Routing:** The vSmart controller is the central point for route exchange and policy enforcement. "The most smartest guy in the SD-WAN solution is this one which is your vSmart." It learns all routes (virtual routes, TLOCs) in the inbound direction via OMP (Overlay Management Protocol) pairings from the edges and then advertises them back.
- vManage: The Administrator's Control Panel:** While vSmart is smart, the administrator (referred to as "Mr. Capil" in the example) controls vSmart through vManage. "You want to control this guy... there must be something through which you can control vSmart. And that something is your vManage."
- NetConf Protocol:** vManage uses the NetConf protocol over a secure TLS/DTLS connection to instruct vSmart.
- Policy Creation:** All policy definitions (site lists, logic, application) are performed on **vManage**.
- Pushing Policies:** When an administrator applies a policy in vManage, it converts the GUI-based configurations into commands and pushes them to vSmart. "vManage is going to convert all of whatever you have specified in the GUI it is going to convert them into the commands and it is going to send them over the netcon protocol to vSmart."
- Policy Enforcement:** Once vSmart receives and "obeys" the instructions from vManage, it enforces these policies on the OMP pairings with the edge devices. This means vSmart controls "what it is advertising to your complete fabric to your entire SD-WAN."

4. Reading SD-WAN Policies in vManage (CLI Preview)

The speaker demonstrates how to interpret the CLI preview of a centralized SD-WAN policy in vManage. He suggests reading from "bottom to top" for better intuition.

Example Hub-and-Spoke Policy Breakdown (from bottom to top):

- apply policy hub-and-spoke site-list spokes out:** This is the application of the policy.
- apply policy:** Indicates policy application.
- hub-and-spoke:** The name of the policy.
- site-list spokes:** Specifies that the policy applies only to sites defined in the spokes site list.

- **out:** The direction of policy application (outbound from vSmart).
- **site-list spokes site-id 1, 2:** Defines the spokes site list to include site IDs 1 and 2.
- **site-list hub site-id 100:** Defines the hub site list to include site ID 100.
- **prefix-list all-ipv4-prefixes ip-prefix 0.0.0.0/0 less-equal 32:** Defines a prefix list that matches all IPv4 prefixes.
- **control-policy hub-and-spoke:** Declares the start of the "control policy" named hub-and-spoke.
- **sequence 1:** First sequence in the policy.
- **match tloc site-list hub:** Matches TLOCs belonging to the hub site list (e.g., T1).
- **action accept:** Permits these matched TLOCs to be advertised. (This filters out TLOCs of other spokes from being advertised to the spokes).
- **sequence 11:** Second sequence in the policy.
- **match route prefix-list all-ipv4-prefixes:** Matches all virtual routes.
- **action accept set tloc-list T1_tloc:** Accepts all virtual routes but rewrites their next-hop TLOC to T1 (the hub's TLOC). This ensures reachability through the hub.
- **Implicit Deny:** The speaker reminds that like access lists, there's an "implicit deny at the end" for actions not explicitly permitted.

5. Types of SD-WAN Policies: Centralized vs. Localized

SD-WAN policies are broadly categorized by their scope:

- **Centralized Policies: Scope:** Affect the "whole SDN fabric from a central location."
- **Applied On:** Pushed to the **vSmart controller**.
- **Impact:** Changes implemented via centralized policies are communicated to edge devices through OMP updates.
- **Sub-types: Control Policies:** Affect the control plane (routing).
- **Topology Policies:** What was discussed, controlling the full-mesh, hub-and-spoke, or hierarchical topologies by manipulating virtual routes and TLOCs.
- **VPN Membership Policies:** Define specific topologies for different VPNs (e.g., hub-and-spoke for guest VPN, full-mesh for corporate VPN).
- **Data Policies:** Affect the data plane (traffic flow).
- **Application-Aware Routing:** Directs traffic for specific applications based on network performance metrics (latency, jitter, packet loss), potentially steering traffic over preferred links (e.g., MPLS vs. Internet).
- **Localized Policies: Scope:** Applicable to a **single edge device** or a group of specific edges. "It is going to affect that edge where I am pushing it."
- **Applied On:** Pushed directly to the **edge devices** (via vManage, using templates over NetConf).
- **Impact:** Affects only the local behavior of that specific edge.
- **Sub-types: Control Policies (Localized):** Affect routing protocols running locally on an edge (e.g., OSPF, BGP configurations on a branch router).
- **Data Policies (Localized):** Affect local traffic forwarding on an edge (e.g., DSCP marking of packets, blocking traffic on a specific interface based on 6-tuple matching, similar to local access lists).

6. Real-World Design Example: Hierarchical SD-WAN Across Regions

The speaker provides a real-life example of a complex SD-WAN design for a customer with three major geographical regions (Asia-Pac, EMEA, US).

- **Customer Requirement:** The customer desires full-mesh connectivity within each region but *not* across regions (to prevent tunnels between every site in Asia-Pac and every site in the US). They still need inter-regional communication.
- **Design Solution (Hierarchical Design): Site ID Planning:** Assign distinct site ID ranges to each region (e.g., 1-1000 for Asia-Pac, 1001-2000 for EMEA, 2001-3000 for US).
- **Border Routers:** Identify specific "border routers" within each region.
- **Policy Logic:** Use centralized policies to:
 - Allow full-mesh **within** each region (by advertising all TLOCs internally).
 - Restrict cross-region TLOC advertisement to only the "border routers" of each region. This means a regular branch in Asia-Pac will not form a direct tunnel with a regular branch in the US.
 - When a branch in one region wants to communicate with a branch in another, the traffic will first go to its regional border router, then traverse a limited number of inter-regional tunnels formed between border routers, and finally reach the destination branch within the other region.

This complex design is achievable using the control policy framework and manipulating TLOCs and virtual routes, emphasizing the power and flexibility of SD-WAN policies.

Day 27 & 28:

SD-WAN Deployment and Migration Strategies: Key Briefing

Overview: The source provides an in-depth discussion on SD-WAN deployments, with a primary focus on "brownfield" migrations – transforming existing legacy WAN solutions to SD-WAN. The speaker, Vishnu, emphasizes that greenfield deployments (starting from nothing) are rare. The session aims to equip listeners with a comprehensive understanding of migration considerations, simplifying the process, topology choices, and traffic paths during transition. The insights shared are based on the speaker's practical experience in real-life SD-WAN implementation projects.

1. Core Principles of SD-WAN Migration

- **Focus on Brownfield Deployments:** The overwhelming majority of SD-WAN projects involve migrating existing legacy WAN infrastructure. "You are not going to get a green field deployment... you are not going to start deploying SD-WAN absolutely from nothing. There must be something at the customer premises with respect to WAN and they are going to say that you need to improve my existing WAN."
- **SD-WAN is Fundamentally Routing:** Despite its advanced features and controller-based architecture, the core function of SD-WAN remains routing. Understanding existing routing protocols and traffic flows is paramount for successful migration. "If you start thinking SD-WAN that SD-WAN is just the routing then you will be happy and you will be understanding most of the concepts pretty easily at the end of the day what you are doing you're doing routing."
- **Simplify the Migration Process:** Migration windows often occur during off-hours and can be lengthy (10-12 hours, sometimes 2-3 days for huge migrations). Simplifying the process is crucial to ease the burden on network engineers. "Everybody wants to simplify their life... try to use automation wherever possible."
- **Confidence Building for Customers:** Customers are often reluctant to change. It's essential to assure them that existing functionalities will continue to work while highlighting the benefits of the new solution. "You are going to give this confidence to the customer that this technology actually works. Believe me whatever was working previously it is going to be work like this."

2. Key Migration Considerations

1. Controller Deployment (On-Premise vs. Cloud):

- This is the first consideration for any SD-WAN deployment.
- **On-Premise:** Financial and government customers often prefer on-premise deployment of vSmart, vBond, and vManage controllers due to concerns about sending sensitive enterprise information (including routing) outside their network. This requires the customer to manage certificates and infrastructure.
- **Cloud (Cisco Cloud):** Most other sectors (e.g., health, sports) tend to choose Cisco cloud-hosted controllers, with Cisco managing the vSmart, vBond, and vManage. This offloads significant operational responsibility from the customer.
- **Redundancy:** Regardless of deployment model, redundancy is critical. While Cisco has tested up to 20 vSmarts, a minimum of two is common for redundancy.
- **Regional Controllers:** For large, geographically diverse customers (e.g., 400-600 sites per region), it is often suggested that "every region have their own controllers," especially vSmart, to reduce inter-regional TLS/DTLS tunnels, though vSmarts across regions still communicate for consistent routing information. Affinity can be configured to direct branches to regional vSmarts.

1. Traffic Paths and Routing (LAN/WAN Integration):

- Understanding the customer's current traffic flow and routing (both LAN and WAN) is the most important step; failure to do so will lead to migration failure.
- "Understanding the current traffic flow of customer is very very important when you consider the migration. If you do not do this step, your migration is going to fail for sure."
- SD-WAN routers integrate with the customer's LAN environment (e.g., OSPF, EIGRP). The SD-WAN solution takes LAN routes and distributes them efficiently across the WAN.
- **LAN-Side Responsibility:** The configuration of the LAN side of the SD-WAN device (VPN 0 for transport, service-side VPNs like VPN 10 for LAN connectivity) is the migration engineer's responsibility. This includes understanding existing LAN routing protocols and redistributing routes (e.g., EIGRP into OSPF).
- **Preserving Production Traffic:** Customers typically do not want downtime for production traffic, especially in data centers. This often means placing new SD-WAN devices in parallel with existing WAN routers rather than direct replacement. "Customer says right now everything in production... I cannot lose this router even for 12 hours also specifically my data center router."
- **Current Traffic Flows:** Analyze how customers currently access the internet (e.g., backhauling to DC) and internal applications. This understanding allows for suggesting improvements with SD-WAN while ensuring existing functionalities are maintained.

1. Segmentation and Firewall Ports:

- **Segmentation (VRFs):** Understand the customer's LAN-side segmentation (e.g., VRFs) and policy requirements (e.g., full mesh for corporate VRF, hub-and-spoke for guest VRF). SD-WAN can control routing at the VPN level.
- **Firewall Ports:** SD-WAN devices require specific firewall ports (TLS, DTLS) to be open to communicate with controllers (vBond, vManage, vSmart), especially when controllers are in the Cisco cloud. Without this connectivity, edge devices cannot join the SD-WAN fabric.

1. Application-Aware Routing:

- Identify the customer's most critical applications and their SLA parameters.
- During migration, it's advisable to migrate applications one by one, starting with less critical ones, to test functionality and minimize impact.
- Large migrations (e.g., 600 branches) require significant planning (designing the network, strategizing migration) and can take a year or more, migrating sites gradually (one by one initially, then a few at once).

3. Sequence of Migration

1. **Deploy Controllers First:** vManage, vSmart, and vBond should be deployed first, and certificates installed (if on-premise). Define site numbers and management networks.

2. **Deploy Hub/Data Center First:** Migrate the hub (data center) sites before branches.

- **Reason:** The hub acts as a bridge between SD-WAN and non-SD-WAN (legacy) sites during the transition phase. "Deploy your hub and then start thinking about migrating your branches one by one."

3. **Migrate Branches Gradually:** Once hubs are operational, branches can be migrated one by one or in small batches.

4. Simplifying the Migration Process

1. **Cloud-Hosted Management:** Choosing Cisco cloud for controllers significantly reduces the customer's operational burden, as Cisco handles the spinning up and management of controllers. "It is going to ease your life guys. If you choose cloud hosted management."

- **Wise Use of Templates and Centralized Policies/Templates:** "Create one once, use it many times." This applies to feature templates (e.g., OSPF template with variables) and device templates. Avoid creating too many templates, as it becomes difficult to manage.
- **Categorization of Sites:** Categorize customer sites based on common characteristics (e.g., "extra-large" for DCs with 3 transports and two service VPNs, "large" for campuses with 2 transports, "medium" for branches with 1 MPLS, "small" for remote branches with only Internet/LTE). This allows for creating standardized device templates for large numbers of similar sites (e.g., 200 "medium" branches using one template).
- **Zero-Touch Provisioning (ZTP):** ZTP allows for automatic device configuration upon connectivity, significantly simplifying remote branch deployments.
- The vManage pre-stages configurations based on device serial numbers. Devices connect to vBond, learn vManage/vSmart addresses, form tunnels, and receive their configurations via NETCONF.
- **Prerequisite:** Devices must obtain an IP address (preferably via DHCP) and Internet connectivity to reach vBond. If DHCP is unavailable, a "bootstrap configuration" (manual IP, vBond address) is needed, which adds a manual step. "Use zero touch provisioning wherever possible."

5. Topology Considerations

SD-WAN offers flexibility to create various topologies, leveraging the default full-mesh behavior of vSmart or controlling it with centralized policies:

- **Full Mesh:** Default vSmart behavior: collects all routes, advertises them to everyone, resulting in full mesh connectivity.
- Suitable for smaller enterprises (e.g., 100 sites) where branches can directly communicate. "You should go for the full mesh... No issues."
- **Hub and Spoke:** Used when branches should not communicate directly with each other (e.g., 600 branches in a region).
- Configured by programming vSmart to advertise only hub TLOCs to spokes and change the next-hop of virtual routes to the hub's TLOCs.
- **Hierarchical SD-WAN (Large-Scale, Multi-Region):** Analogous to OSPF areas (Area 0 as backbone, other areas as non-backbone).
- Addresses the need for full mesh within regions but not between regions (e.g., to avoid intercontinental tunnels between all branched).
- **Concept:** Designates "border routers" within each region that form inter-regional tunnels (similar to ABRs in OSPF Area 0). Other branches within a region communicate via their regional hub/border router to reach other regions.
- Achieved by controlling TLOC advertisement (not advertising TLOCs of other region's branches) and manipulating next-hops of virtual routes to point to regional hubs/border routers.
- Requires careful site ID planning to logically group regions and border routers.

6. Data Center and Branch Migration Placement

1. Data Center Placement (Parallel Deployment):

- Typically, new SD-WAN edge routers are placed in parallel with existing legacy WAN routers to avoid downtime.
- The SD-WAN edge connects to the existing LAN switch (e.g., corporate LAN VPN 10) and the existing WAN transport (VPN 0).
- Routes from the LAN are learned by the SD-WAN edge (e.g., OSPF, EIGRP), advertised to vSmart, and then distributed to other SD-WAN sites.
- TLOC extension can be used to provide redundancy or multiple transports to the SD-WAN edge device.
- Flexibility in connection: SD-WAN edges can connect directly to distribution switches, extending VLANs to connect LAN and WAN sides.

1. Branch Migration Placement:

- **Option 1: In-Place Conversion:** If the existing branch router is compatible with SD-WAN (e.g., ASR 1000 FX series running IOS XE), its operating system can be upgraded, and it can be converted into an SD-WAN router. This eliminates the need for new hardware.
- **Option 2: Parallel Deployment:** If the existing router is not SD-WAN capable or the customer wants no downtime, a new SD-WAN edge router is placed in parallel. It connects to the existing LAN switch and the WAN transport.

7. Traffic Path During Migration (SD-WAN to Legacy Communication)

During a brownfield migration, there will be a mixed environment of SD-WAN and legacy sites. Seamless communication between these is critical.

- **SD-WAN Transformed Site to SD-WAN Transformed Site:** SD-WAN edges learn LAN routes, advertise them to vSmart via OMP.
- vSmart distributes these routes and TLOCs to other SD-WAN edges.
- SD-WAN edges establish IPsec tunnels based on TLOC information.
- Traffic flows directly over these secure tunnels between SD-WAN sites.
- **SD-WAN Transformed Site to Legacy Site:** **Legacy Site to SD-WAN Site (e.g., Legacy Branch to SD-WAN DC):** LAN routes from the SD-WAN transformed site (e.g., Data Center) are learned by the SD-WAN edge.
- These routes are then redistributed from OMP into the local LAN routing protocol (e.g., OSPF) that the legacy router understands.
- The legacy router learns this route to the SD-WAN transformed site.
- Traffic from the legacy site flows through its existing WAN connection to the SD-WAN edge at the destination.
- **SD-WAN Site to Legacy Site (e.g., SD-WAN Branch to Legacy Branch):** Legacy branch LAN routes are learned by its existing WAN router.
- This route is advertised into the MPLS/WAN cloud and is learned by the SD-WAN data center edge.
- The SD-WAN data center edge redistributes this route into OMP.
- vSmart advertises this route to the SD-WAN transformed branch.
- Traffic from the SD-WAN branch flows over its SD-WAN tunnel to the data center, then through the data center's legacy WAN connection to the legacy branch.
- **Key takeaway:** "It is just plain routing." The challenge is ensuring routes are properly redistributed between OMP and traditional routing protocols (OSPF, EIGRP) at the points of intersection (e.g., where SD-WAN edges connect to legacy LAN/WAN segments). The goal is to make the traffic flow appear seamless to end-users who are unaware of the underlying WAN transformation.

Conclusion: Successful SD-WAN migration, particularly in brownfield scenarios, relies on meticulous planning, a deep understanding of existing network infrastructure (especially routing and traffic flows), and strategic deployment of SD-WAN controllers and edge devices. Leveraging automation, templates, and centralized policies significantly simplifies the process. The ability to manage communication between legacy and SD-WAN components during the transition is critical.

Day 29 & 30:

Detailed Briefing Document: SD-WAN Service Chaining and Cloud OnRamp Solutions

1. Session Overview

This session, the 29th and 30th hour of the course (Session 15th, 2 hours per session), is the final class but marks the beginning of a continuous learning journey with future SD-WAN video uploads and bi-weekly doubt-clearing sessions. The agenda for this session includes:

- **Service Chaining:** An important concept often requested by customers.
- **Cloud OnRamp Solution for SaaS:** Specifically discussing applications like Office 365, Zoom, Webex, and Gmail.
- **Cloud OnRamp Solution for IaaS:** Covering concepts like Virtual Private Clouds (VPCs), regions, and availability zones.

The speaker emphasizes that prior cloud knowledge is not strictly necessary for this session but encourages future exploration of his "cloud from scratch" series for a deeper understanding of cloud networking from an engineer's perspective.

2. Service Chaining

Service chaining allows an SD-WAN solution to intelligently redirect traffic through specific network services, such as firewalls, Intrusion Prevention Systems (IPS), Intrusion Detection Systems (IDS), or load balancers, typically located in a data center, before forwarding it to its final destination.

A. Core Concept: VPN Labels and Traffic Redirection

- **VPNs and Labels:** The SD-WAN solution assigns a unique label to each VPN (e.g., VPN 10 assigned label 3121, VPN 20 assigned 3122) at the service or LAN side of an edge device (VEdge/VH). These labels are locally scoped.
- **Advertisement via OMF:** VEdges advertise their service-side routes, along with their corresponding VPN labels and VPN numbers, to the VSmart controller via the Overlay Management Protocol (OMF).
- **Data Plane Behavior:** When traffic intended for a specific VPN (e.g., VPN 10) leaves a VEdge, its IPsec header includes the associated VPN label (e.g., 3121). The destination VEdge uses this label to quickly identify which VRF (or VPN) the packet belongs to, allowing for direct forwarding without extensive routing table lookups.
- **Problem Statement:** While traditional routing can achieve similar outcomes, customers often desire a simplified, "SD-WAN in a simplified manner" approach for traffic inspection, especially for hub-and-spoke topologies where all inter-branch traffic must pass through a centralized data center for security scrutiny (e.g., firewall inspection).

B. How Service Chaining Works

1. Service Advertisement:

- The VEdge directly connected to the service (e.g., a firewall) is configured to advertise this service to the VSmart controller via OMF.
- This advertisement includes a unique VPN label for the service itself, distinct from the VPNs that traverse it. For example, if VPN 1 has label 3170, a firewall connected to VPN 1 might be assigned label 3171.
- "What is the meaning of services? The meaning is whether you have configured any firewall here. Whether you have configured any IPS here IDS here maybe load balancer here. Right. What what These things are all these are services because in your data center you can have firewall, you can have IPS, IDS, load balancer, so many things, right?"

1. Centralized Policy Configuration:

- The network administrator defines centralized policies on the VSmart controller (via VManage).
- These policies dictate that specific traffic (e.g., communication between Branch 1 and Branch 2) should have its "next hop" set to the advertised service (e.g., "set next as firewall").
- This "set next hop as firewall" command translates to changing the VPN label in the IPsec header of the traffic from the standard VPN label (e.g., 3170) to the service-specific label (e.g., 3171).

1. Traffic Flow and Inspection:

- When traffic arrives at the VEdge connected to the service (e.g., VH1 in the DC), it inspects the VPN label.
- If the label is the service label (e.g., 3171), the VEdge "is directly going to see that okay 3171 means something different to me it means that I have to switch this packet directly to the firewall and I'm not going to look into the routing table of whatever is in the VPN one."
- After inspection by the firewall, the traffic is forwarded back to the VEdge and then to its final destination.
- The VEdge and the service (firewall) must be directly or Layer 2 connected.

C. Use Cases and Advantages

- **Centralized Security:** Enables organizations to keep their established security policies and high-end security devices (firewalls, IPS, IDS) in the data-center, rather than distributing them to every branch, addressing reluctance from "old school" IT and limitations of lower-end branch devices.
- **Selective Inspection:** Policies can be granularly applied, allowing only specific branches or specific types of traffic to be inspected by the service. "You can say that whenever maybe you have 10 branches right and you do not want to have the service inspection for all the branches. Maybe you select only three branches that whenever the traffic comes from these three branches whenever they are going to talk with each other then they are going to inspect right on this firewall. You can do that."
- **Automation:** The configuration is simplified, with VManage generating the necessary commands. The process of defining Xts, logic, and applying the policy is similar to creating other centralized policies.

3. Cloud OnRamp for Software as a Service (SaaS)

Cloud OnRamp for SaaS optimizes connectivity to public SaaS applications (e.g., Office 365, Zoom, Webex, Gmail) by dynamically selecting the best path based on real-time application quality experience (AQE).

A. Problem Statement

- **Traditional SaaS Access:** Typically, branch users access SaaS applications by traversing IPsec tunnels to a data center, which then connects to the cloud (e.g., via Direct Connect). This adds unnecessary latency: "unnecessarily you are going first to the data center and then basically you are going into the cloud."
- **Direct Internet Access (DIA) Limitations:** While DIA can route specific application traffic directly from the branch to the internet (and then to the cloud), it lacks intelligence to determine the best direct path among multiple ISPs or dynamically react to performance fluctuations.
- **Multiple Paths:** There are often multiple paths to reach SaaS applications: direct via ISP1, direct via ISP2, via tunnel 1 to DC to cloud, or via tunnel 2 to DC to cloud. Determining the consistently best path is challenging.

B. Cloud OnRamp Solution

- **Dynamic Path Selection:** The SD-WAN solution continuously calculates AQE values (Application Quality Experience) for all possible paths to the SaaS application.
- **AQE Metrics:** AQE is determined based on real-time delay, jitter, and packet loss, which are critical for application performance.
- **System-Defined Requirements:** The SD-WAN system inherently knows the optimal delay, jitter, and loss requirements for common SaaS applications like Office 365. It automatically assesses whether a path meets these requirements.
- **Scoring (1-10):** The system assigns an AQE score (1-10, higher is better) to each path.
- **Automatic Path Shift:** "It is the system responsibility to calculate this AQE value... the solution knows that what are the corresponding value of delay jitter loss which are going to work with Office 365." If a path's AQE degrades, the system automatically shifts traffic to the currently best-performing path, without manual intervention.

C. How AQE is Calculated

- **BFD for Tunnels:** For paths traversing IPsec tunnels (e.g., branch to DC), Bidirectional Forwarding Detection (BFD) packets are used to calculate delay, jitter, and loss.
- **HTTP Ping for Direct Paths:** For direct internet paths to the cloud or from the DC edge to the cloud, HTTP ping is used to measure roundtrip time, delay, jitter, and loss.
- **Combined Metrics:** The total AQE for a path might combine BFD metrics (tunnel segment) and HTTP ping metrics (cloud segment).
- **Branch-Specific Calculation:** AQE values are calculated *per branch* and *per application*, meaning different branches might use different paths to the same SaaS application based on their local network conditions. "It doesn't matter the traffic path could be different in different branches based on whatever the application quality experience you have."

D. Configuration

- Network administrators simply define which branches are "client sites" and for which specific SaaS applications they want to enable Cloud OnRamp. The system handles all path calculation and optimization.

4. Cloud OnRamp for Infrastructure as a Service (IaaS)

Cloud OnRamp for IaaS allows an enterprise to seamlessly extend its SD-WAN fabric into public cloud environments (AWS, Azure, GCP) to integrate their own virtual machines and networks hosted in the cloud.

A. Core Cloud Concepts Explained

- **Physical Servers & Virtualization:** Cloud providers utilize vast physical server farms with a virtualization layer to offer virtual machines (VMs) as a service.
- **Regions:** Cloud platforms are geographically divided into regions (e.g., Asia Pacific, Europe, Americas) to reduce latency for users.
- **Availability Zones (AZs):** Within a region, there are multiple, isolated data centers called availability zones. These provide redundancy; users often deploy VMs across different AZs for high availability.
- **Infrastructure as a Service (IaaS):** Unlike SaaS (where the provider manages everything), IaaS gives users control over the virtualized infrastructure (VMs, networking). Users are responsible for configuring their network within the cloud, similar to an on-prem data center.
- **Virtual Private Cloud (VPC):** A VPC (or VNet in Azure) is a logically isolated private network within a public cloud, dedicated to a single customer. It allows users to define their own IP address ranges, subnets, gateways, and virtual firewalls. "whatever the network you are creating inside a particular cloud, whatever the network you are creating that is totally belongs to you, right? That is why it is a cloud inside that cloud. But that is totally dedicated to you. That is your private cloud right and everything is virtual that is why we are calling it the virtual private cloud."
- **VPC Peering:** Allows direct network connectivity between two VPCs. However, a key limitation is that VPC peering is not transitive; VPC1 cannot be used to route traffic between VPC2 and VPC3. This leads to a "full mesh" peering problem for many VPCs.
- **Transit VPC/Gateway:** A special VPC or dedicated service (like AWS Transit Gateway) designed to act as a central hub for connecting multiple other VPCs, solving the peering scalability issue. This enables a hub-and-spoke model within the cloud, allowing VPCs to communicate with each other via the transit VPC.

B. Problem Statement for SD-WAN Integration

- **Branch to Cloud Connectivity:** Traditionally, branches access servers in the cloud by first routing traffic to the on-prem data center, which then connects to the cloud (e.g., via Direct Connect/Express Route). This adds latency. "Why we are going to DC first to access the cloud. Why this branch cannot access this cloud directly?"
- **Integrating Cloud into SD-WAN Fabric:** The goal is to make the cloud environment (VPCs, servers) a seamless part of the SD-WAN fabric, allowing branches to establish direct IPsec tunnels to SD-WAN routers deployed in the cloud.

C. Cloud OnRamp for IaaS Solution

- **Automated Cloud Integration:** VManage automates the deployment of SD-WAN routers (Cisco Cloud Services Routers - CSRs or other VEdges) directly into the customer's cloud environment.
- **VManage's Role:**
 1. **Cloud Credentials:** The administrator provides VManage with cloud provider credentials (AWS, Azure, GCP).
 2. **VPC Discovery:** VManage logs into the cloud console to discover all existing VPCs and their associated VMs.
 3. **Transit VPC/Gateway Deployment:** VManage automatically provisions a transit VPC (if not already present) and deploys SD-WAN routers (CSRs) within it.
 4. **Connectivity Establishment:** VManage automatically connects the customer's existing VPCs to this newly deployed transit VPC/SD-WAN router. This effectively makes the cloud side a "service side" with SD-WAN routers.
 5. **SD-WAN Fabric Integration:** The deployed cloud SD-WAN routers establish DTLS/TLS connections with the VSmart controller and form IPsec tunnels with branch VEdges.
 6. **Configuration Push:** VManage pushes the necessary configurations, bringing the cloud environment into the SD-WAN fabric.
- **Benefits:**
 - **Direct Cloud Access:** Branches can directly establish tunnels to their applications and servers hosted in the cloud, bypassing the on-prem data center for cloud-bound traffic. "this branch can make IPsec connection the SD wan data tunnels to this guy and we can we can talk directly."
 - **Automation:** The entire process of deploying and integrating cloud networking infrastructure and SD-WAN routers is automated, simplifying complex cloud network setups. "The whole soul pro process is automated for you you do not need to because most of the guys doesn't know about it what is transit VPC how you are going to create these connection this connection and that is why they are doing the automation for you."
 - **Unified Fabric:** The cloud becomes "just like any other data center or any other site," seamlessly integrated into the enterprise's SD-WAN network.

5. "OnRamp" Terminology (Clarification Attempted by Participants)

The speaker initially doesn't know the exact origin of the "OnRamp" terminology but allows participants to share their interpretations. The consensus is that "on-ramp" signifies a mode of access or a connection point, similar to an on-ramp connecting a local road to a highway. In the context of cloud, it represents the access point for loading data or resources onto the cloud infrastructure.

6. Q&A and Future Learning

The session concludes with an open Q&A covering various aspects, including:

- **Service Chaining Flexibility:** Confirming that specific VPNs can be selected for inspection.
- **IPsec and Multicast:** Discussion on IPsec supporting multicast traffic (affirmed by a participant).
- **Cloud OnRamp Applicability:** Confirmed to be suitable for Azure and GCP, with terminology differences (e.g., VNet in Azure is equivalent to VPC).
- **Routing in Cloud VPCs:** Discussion on how routing tables in VPCs learn routes, especially concerning transit gateways.
- **Industry Best Practices:** Clarification that for Office 365, the "best practice" is to use whatever path the AQE calculation identifies as best at that moment, rather than a fixed path.
- **Future Courses:** Mentions upcoming courses on BGP/MPLS, Network Security, Network Interview Preparation (including a deep dive into DNS), and SD-Access (LISP).

The speaker expresses gratitude to participants for their questions, highlighting how they enrich the learning experience and build the course content.

